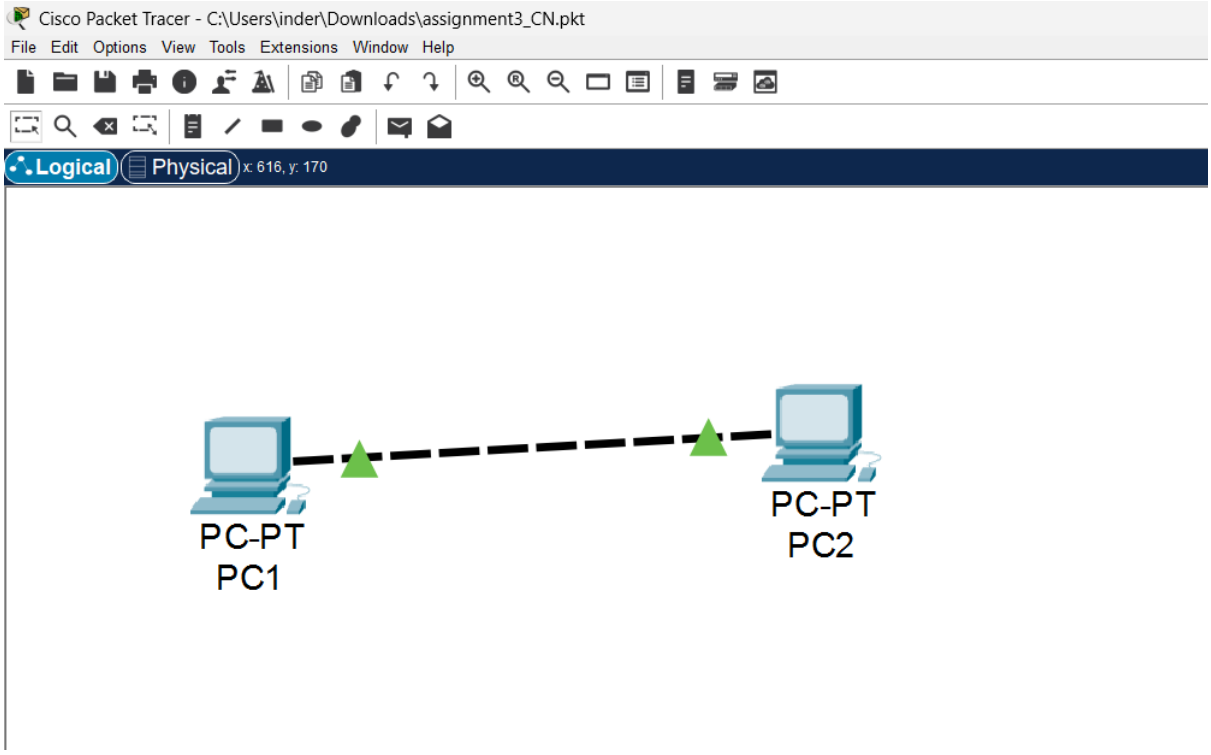
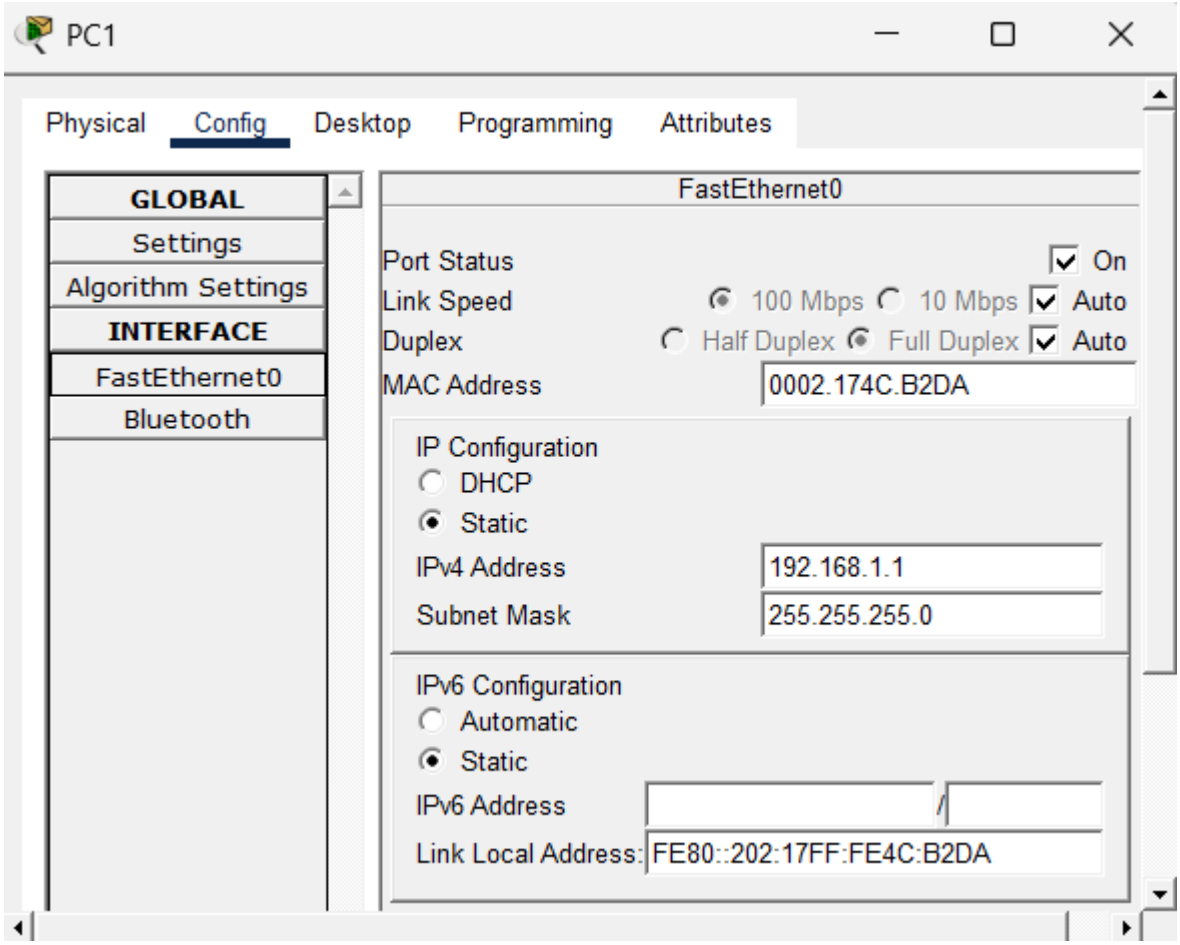
LAB: Assignment 3

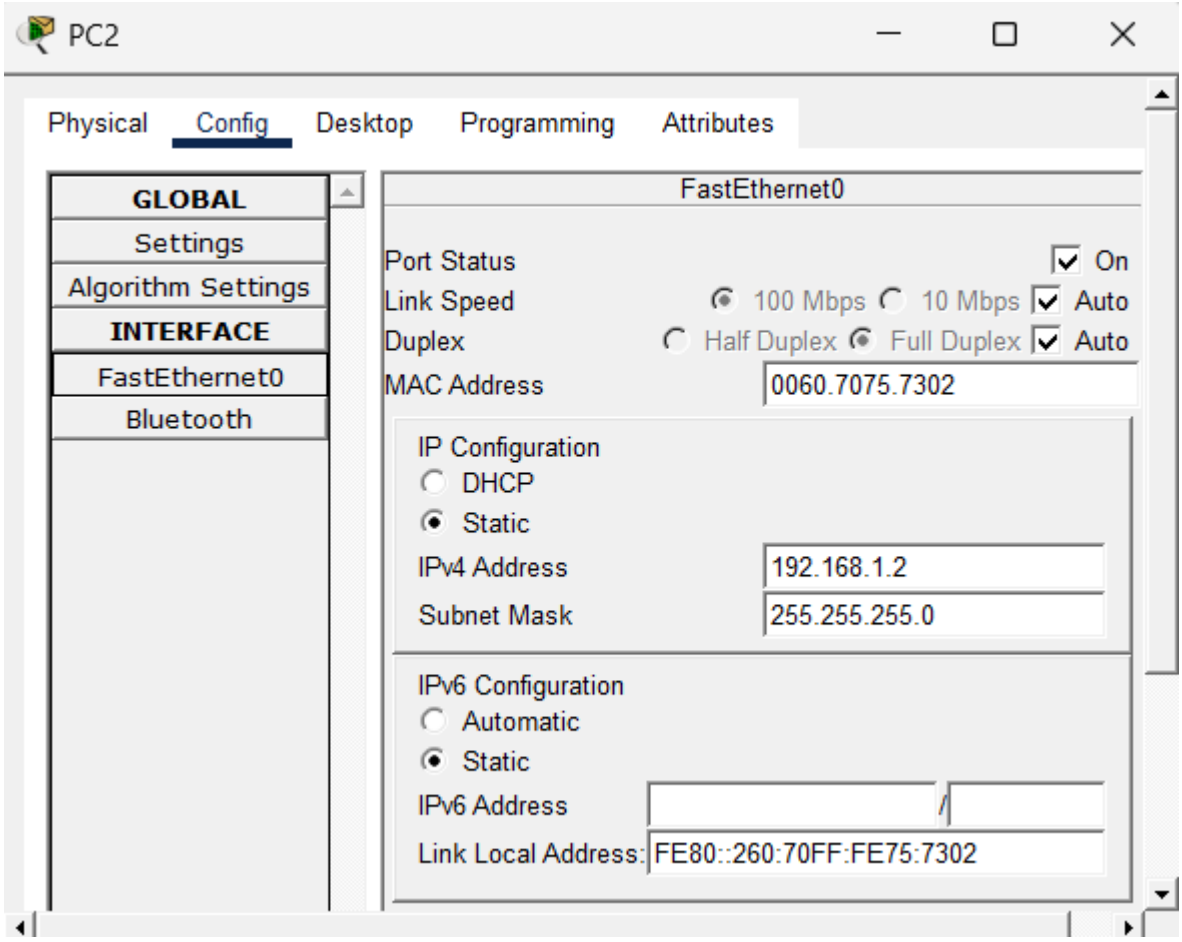
Objective: To create simple Network Topologies Using Cisco Packet Tracer.

Instructions: The instructor is required to discuss the following questions with the students. Students are required to make solution for each question and check connectivity using PING command.

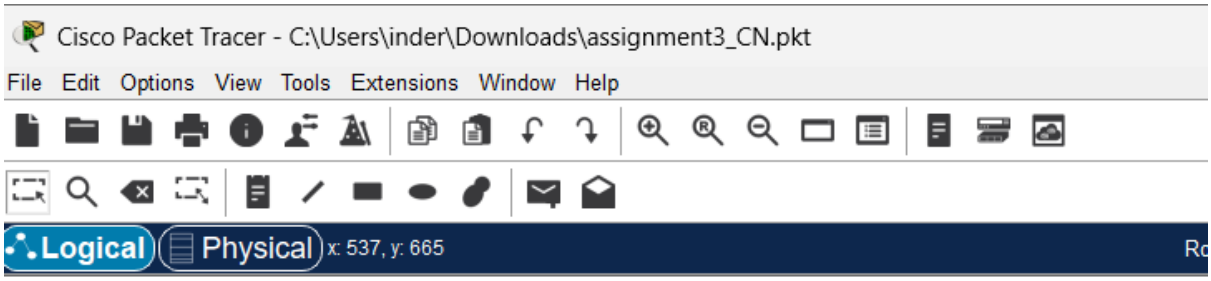
Q1. Connect one PC with another PC as per diagram below.

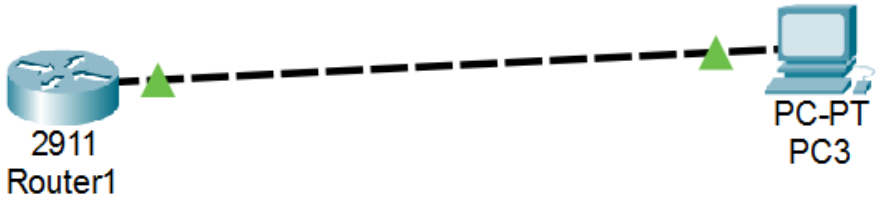


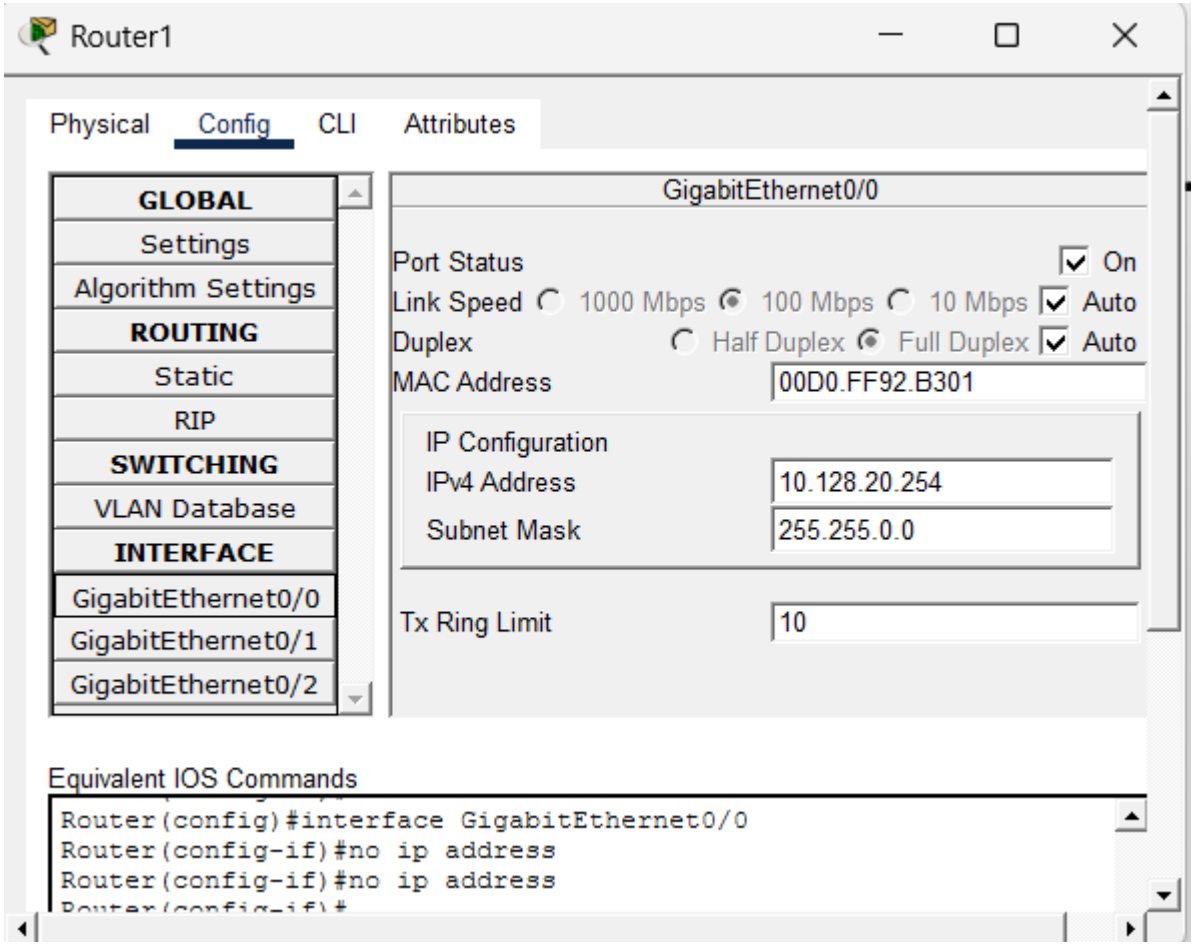

PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0002.174C.B2DA
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::202:17FF:FE4C:B2DA

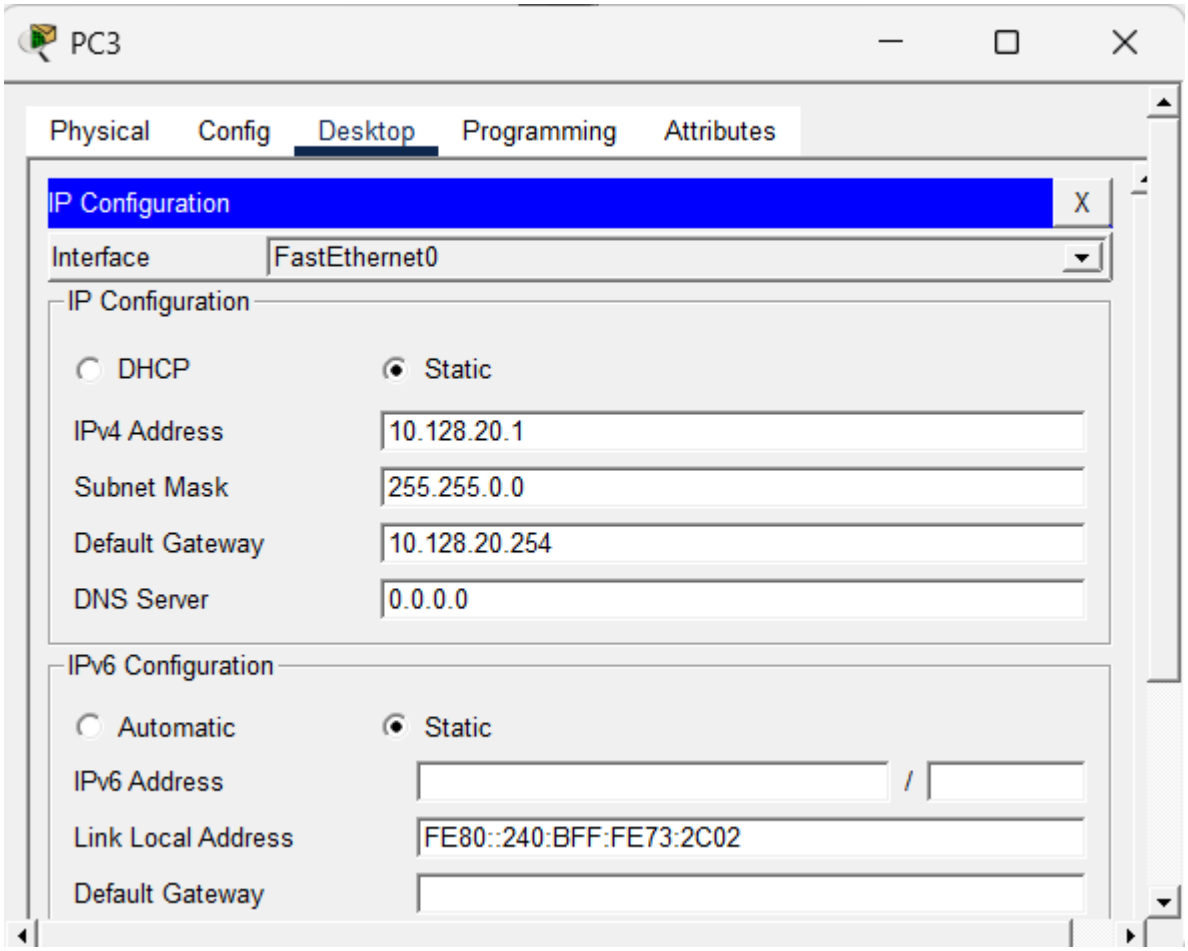

PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.7075.7302
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::260:70FF:FE75:7302

Q2. Connect a PC with Router Ethernet port as shown below.

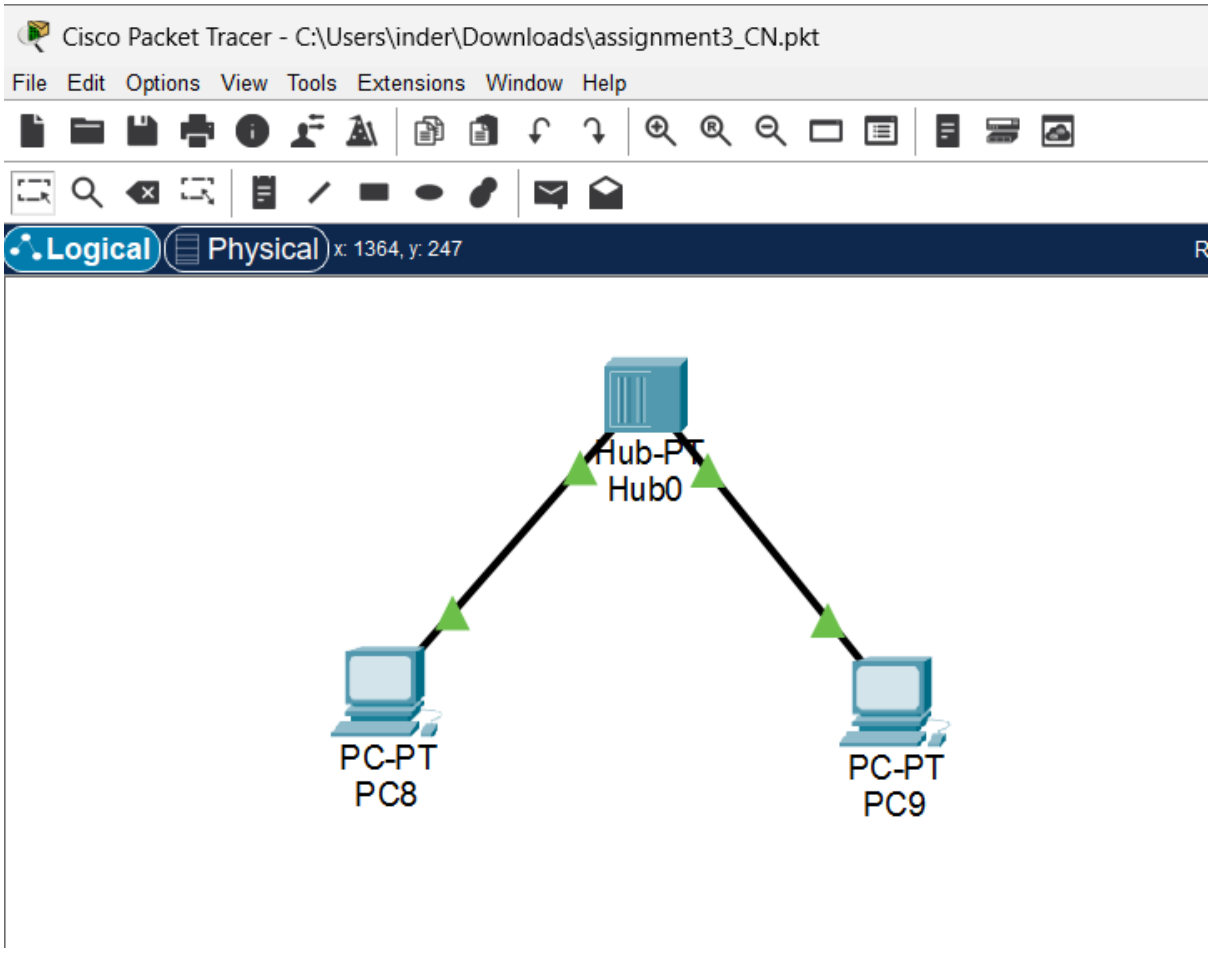


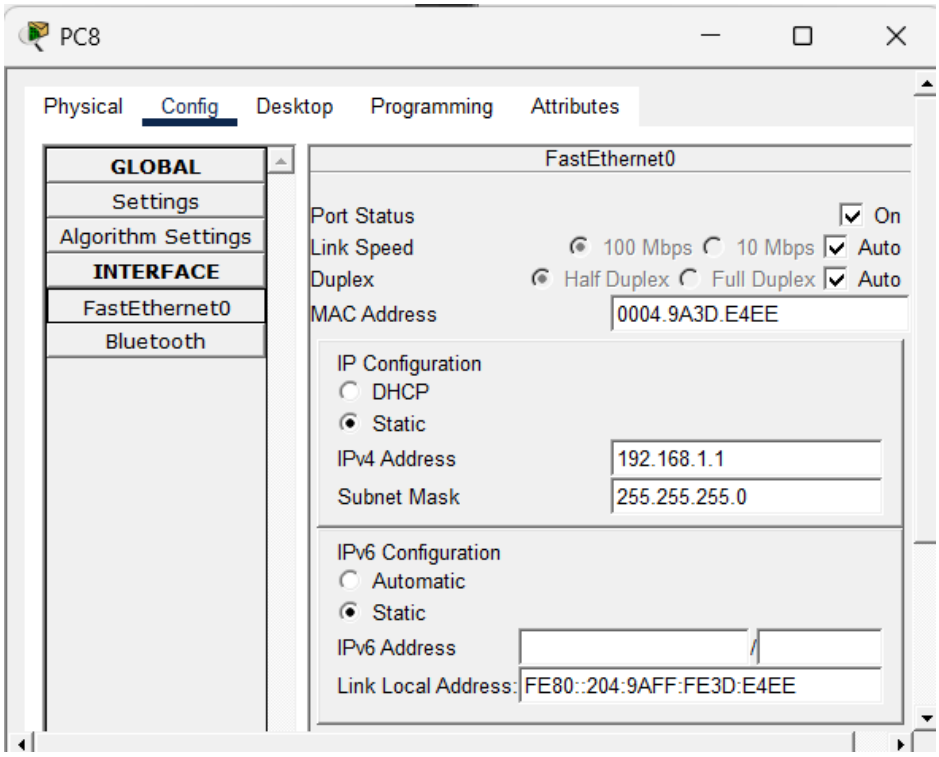


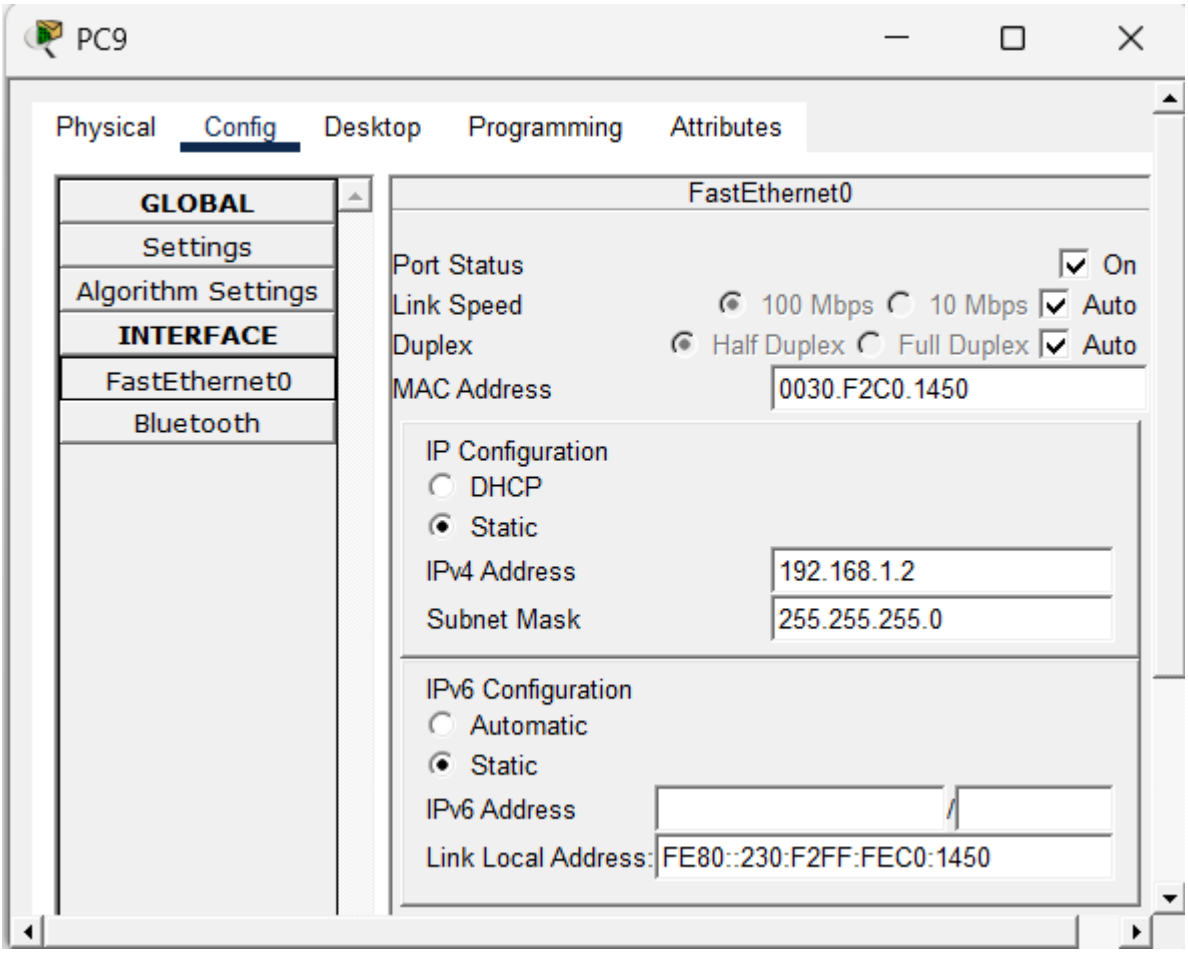



PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.128.20.1
Subnet Mask
255.255.0.0
Default Gateway
10.128.20.254
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::240:BFF:FE73:2C02
Default Gateway

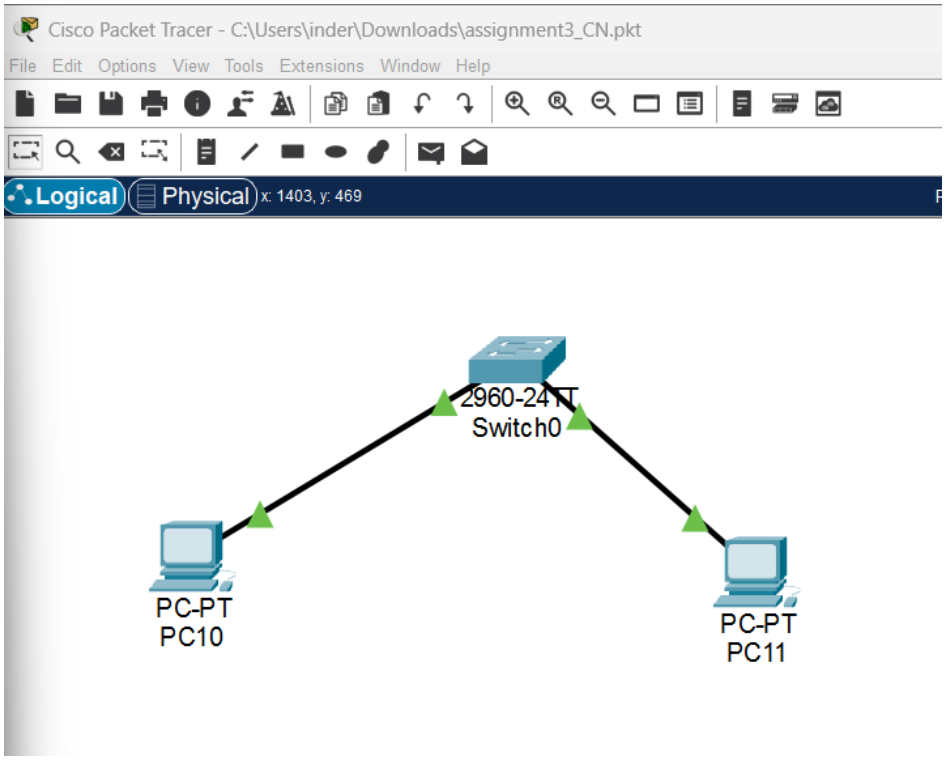
Q3. Create a network of 2 PCs as below.

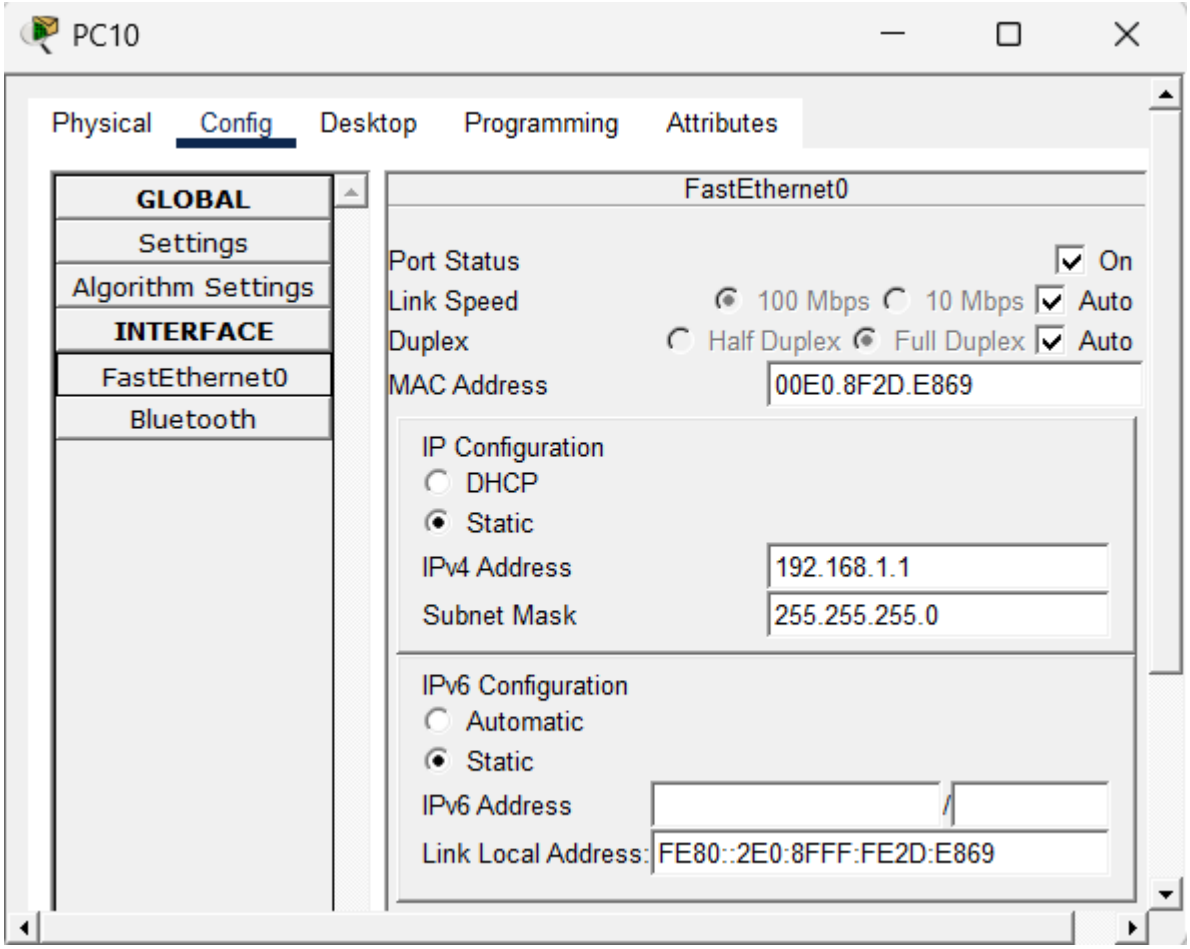


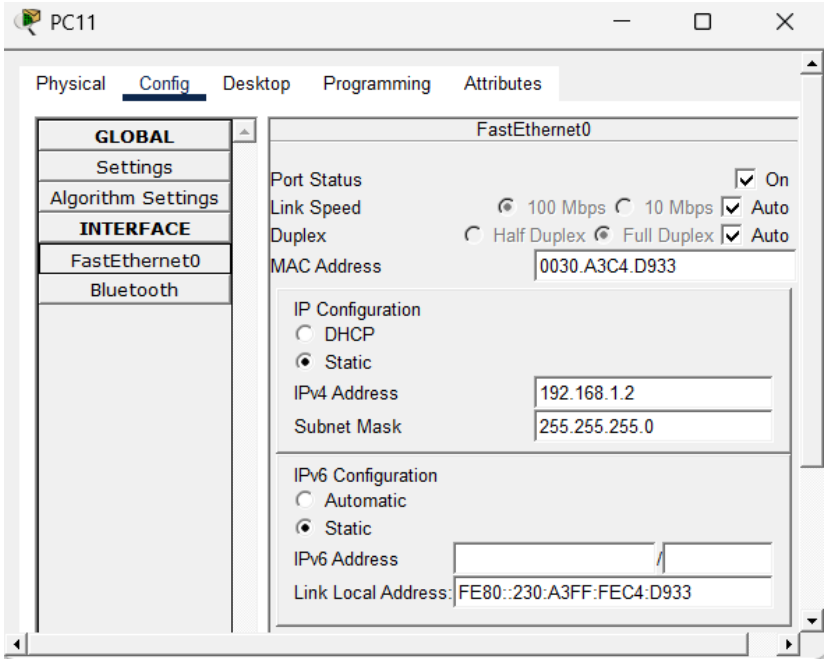


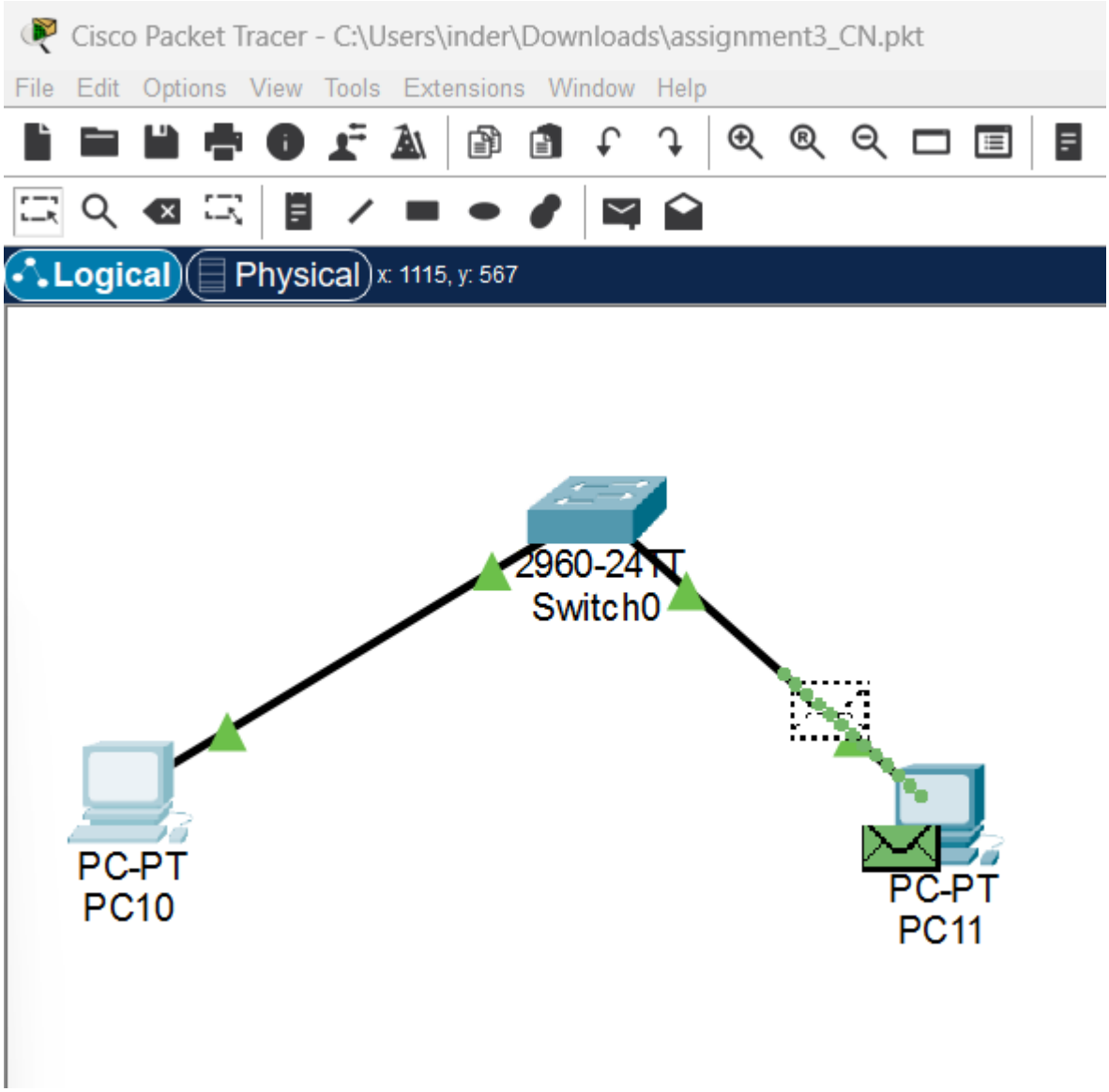

PC9
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F2C0.1450
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::230:F2FF:FEC0:1450

Q4. Connect 2 PCs with a Switch below.



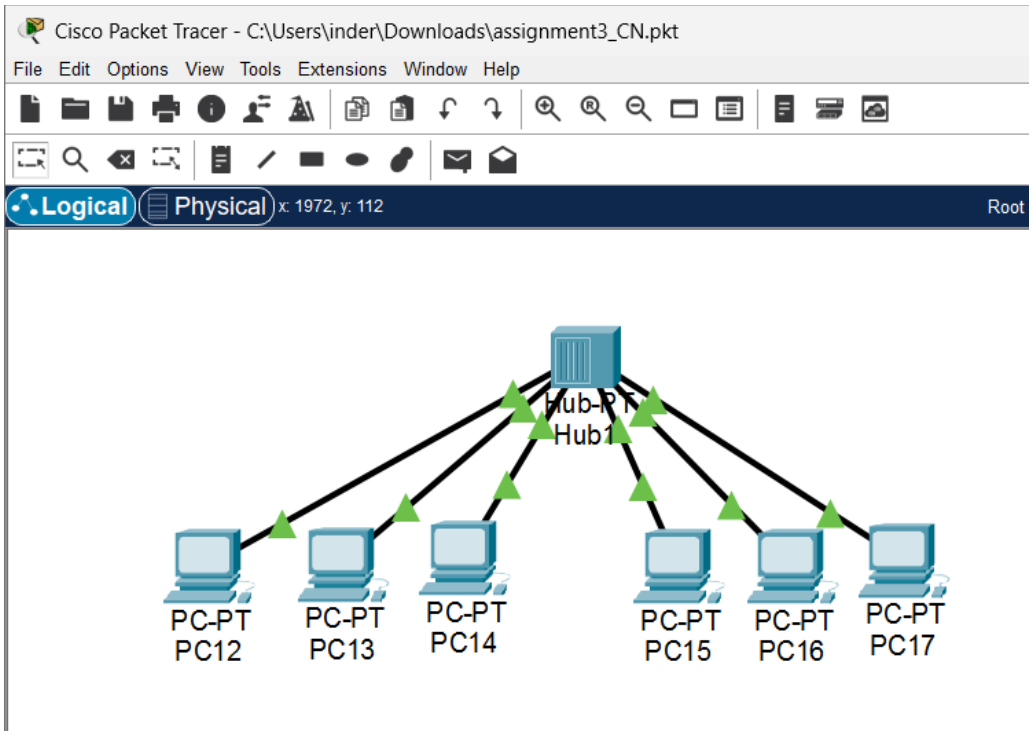


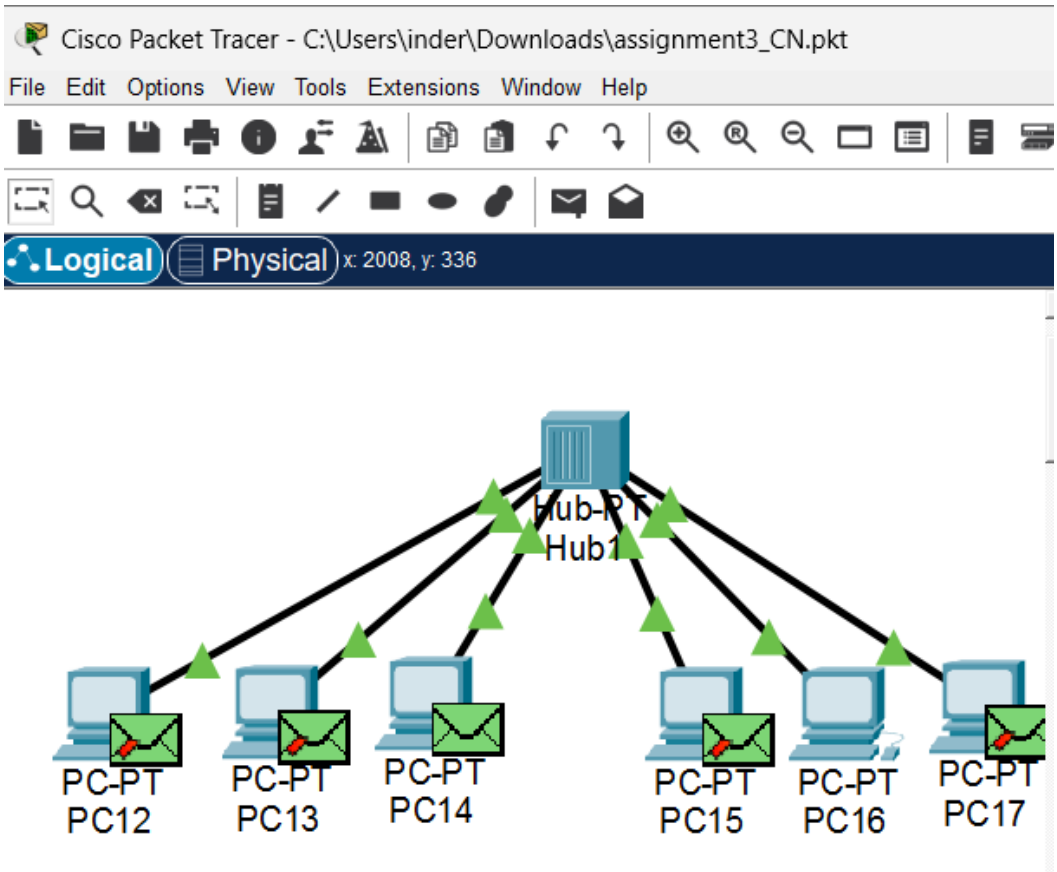

PC11
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.A3C4.D933
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::230:A3FF:FEC4:D933

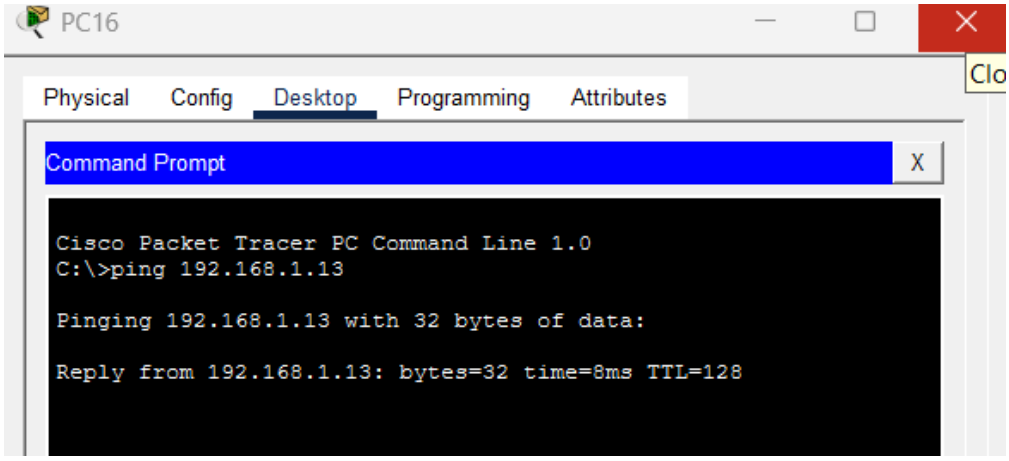

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 1115, y: 567
2960-24TT
Switch0
PC-PT
PC10
PC-PT
PC11

Q5. Create BUS topologies as below.

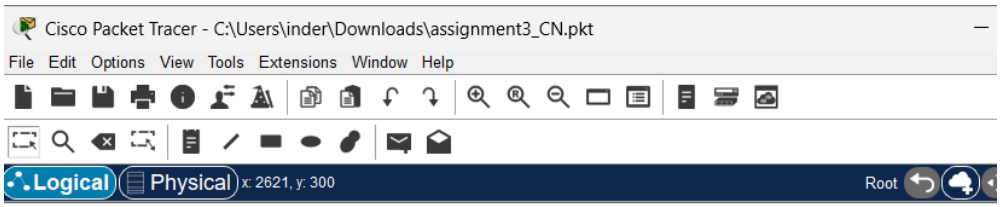
a.

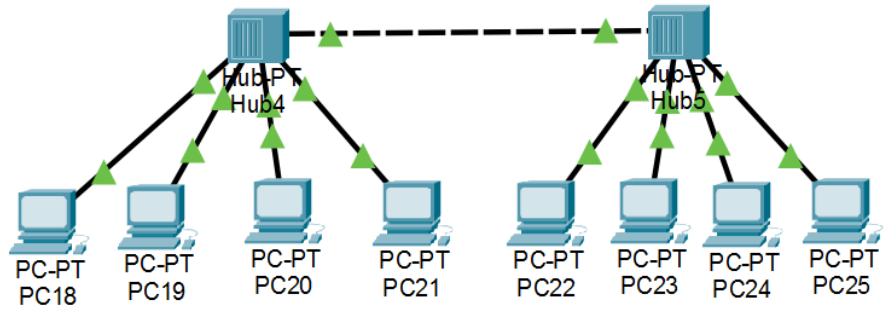


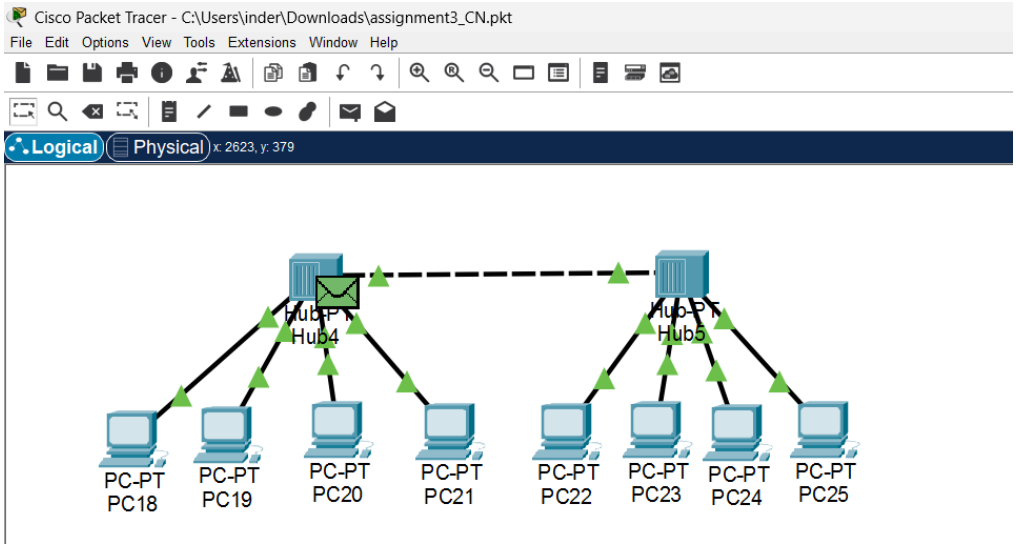


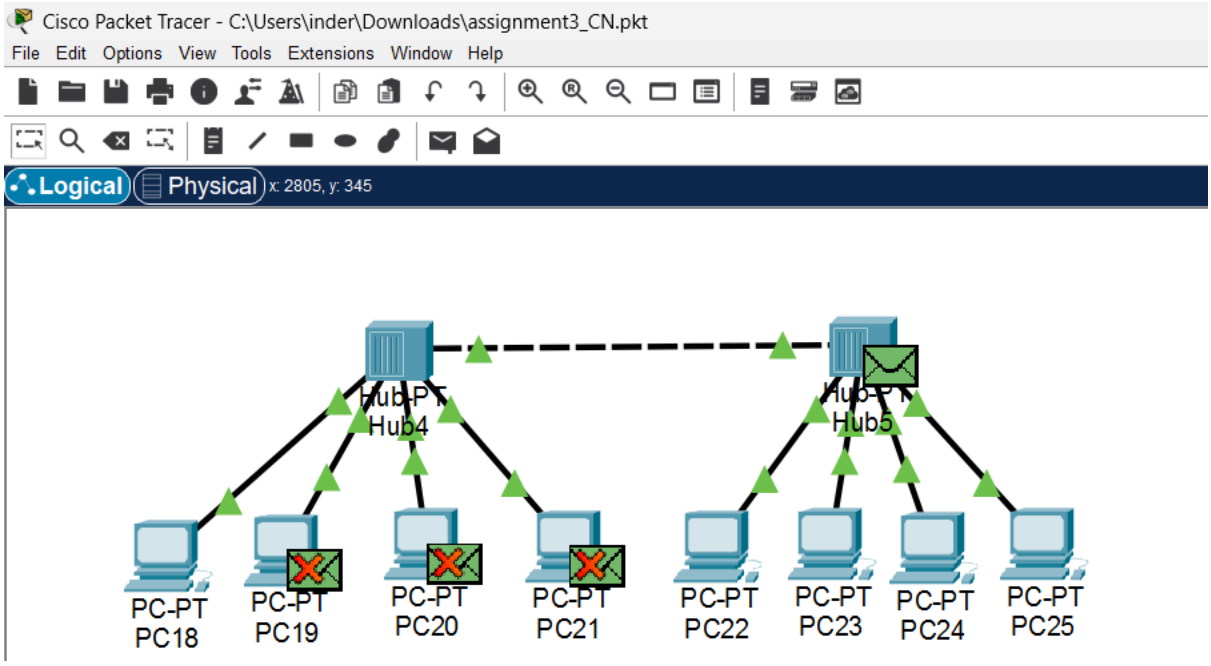


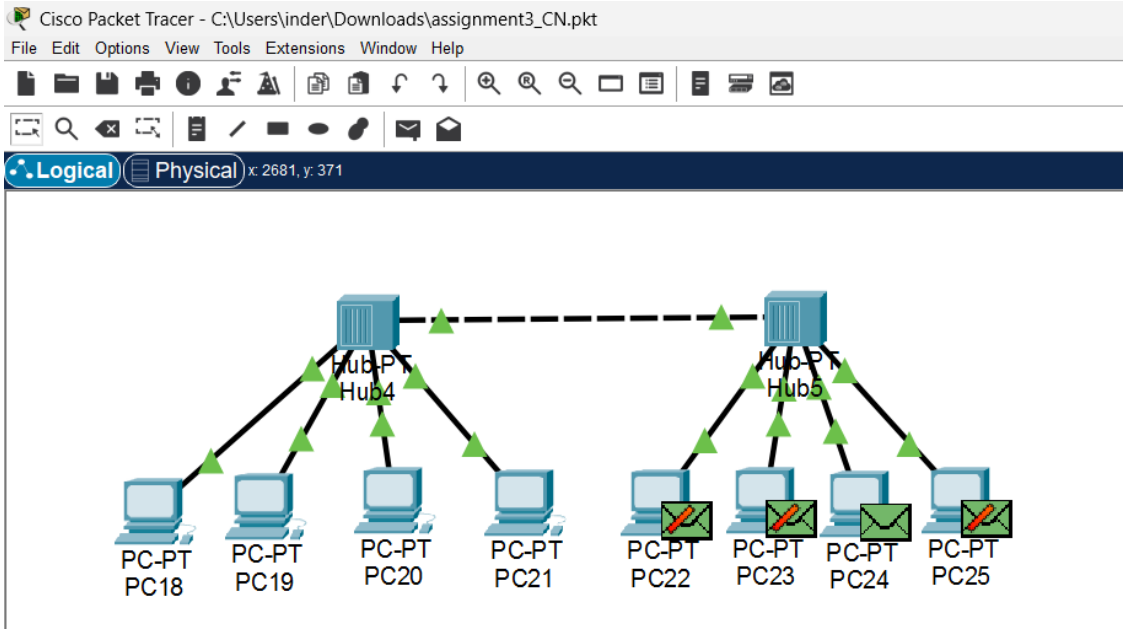
b. Pinging PC24 from PC18

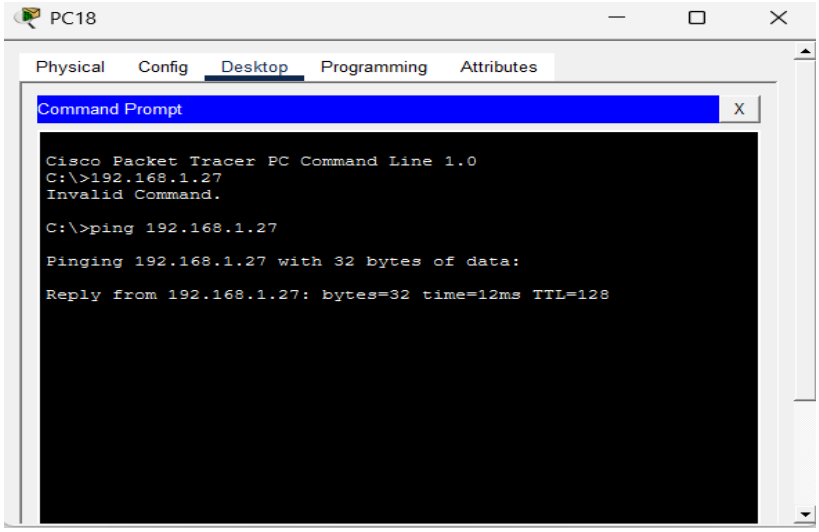




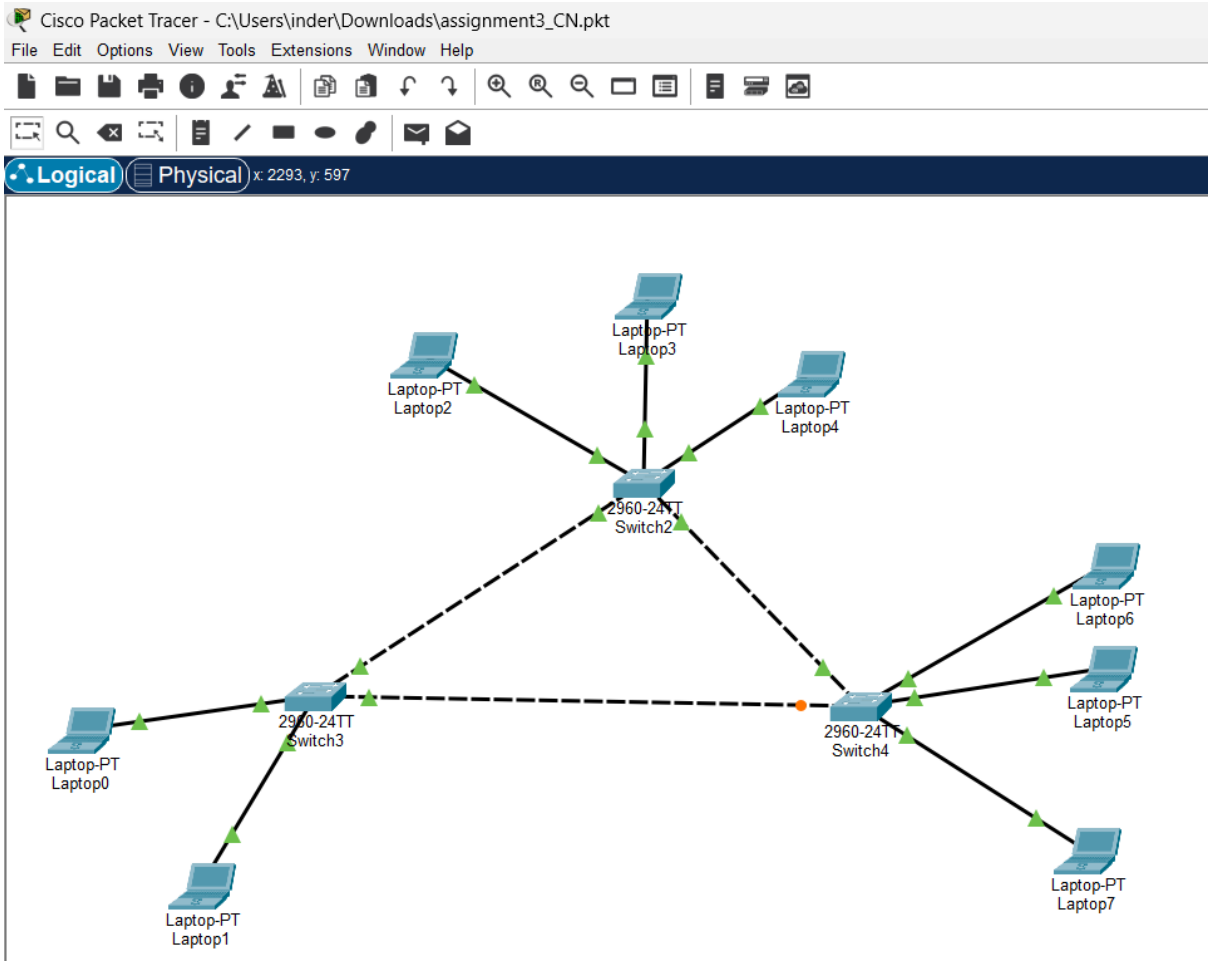



Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2805, y: 345
Hub-PT
Hub4
Hub-PT
Hub5
PC-PT
PC18
PC-PT
PC19
PC-PT
PC20
PC-PT
PC21
PC-PT
PC22
PC-PT
PC23
PC-PT
PC24
PC-PT
PC25

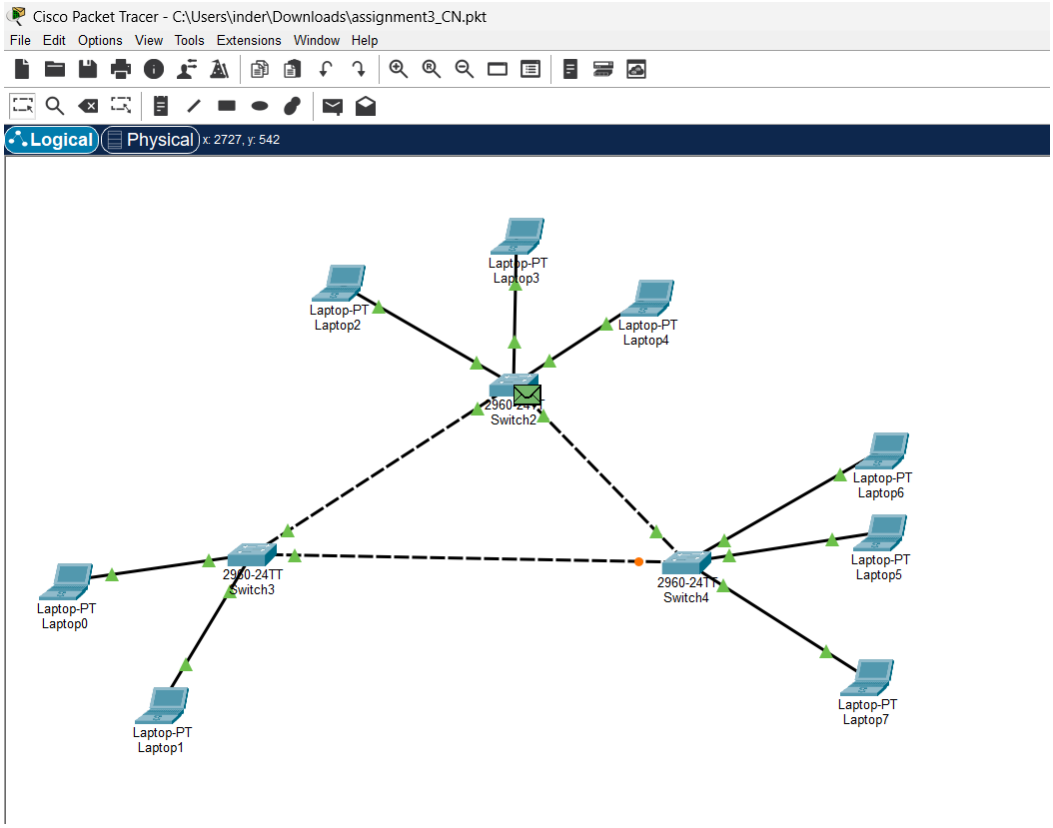

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2681, y: 371
Hub-PT
Hub4
Hub-PT
Hub5
PC-PT
PC18
PC-PT
PC19
PC-PT
PC20
PC-PT
PC21
PC-PT
PC22
PC-PT
PC23
PC-PT
PC24
PC-PT
PC25

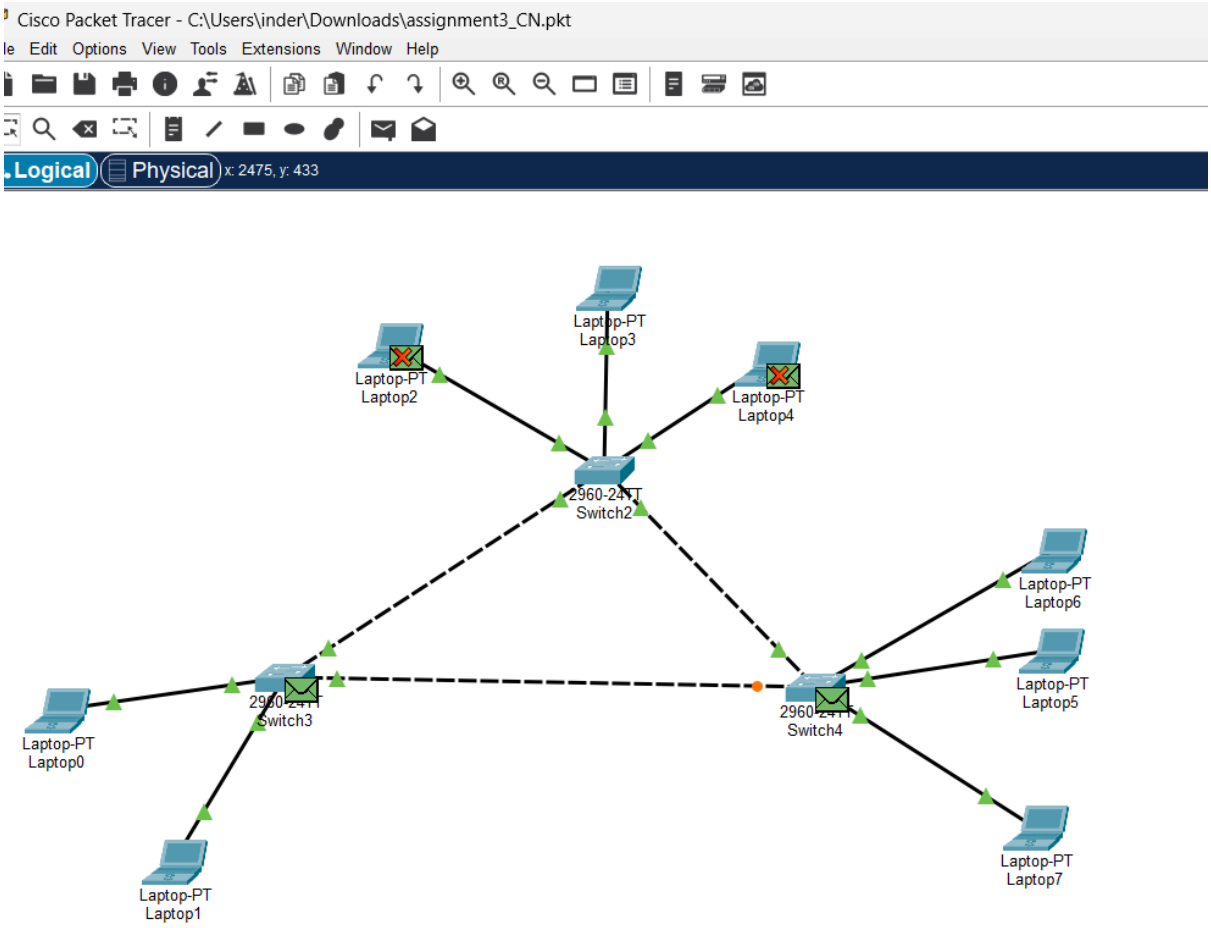

PC18
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>192.168.1.27
Invalid Command.
C:\>ping 192.168.1.27
Pinging 192.168.1.27 with 32 bytes of data:
Reply from 192.168.1.27: bytes=32 time=12ms TTL=128

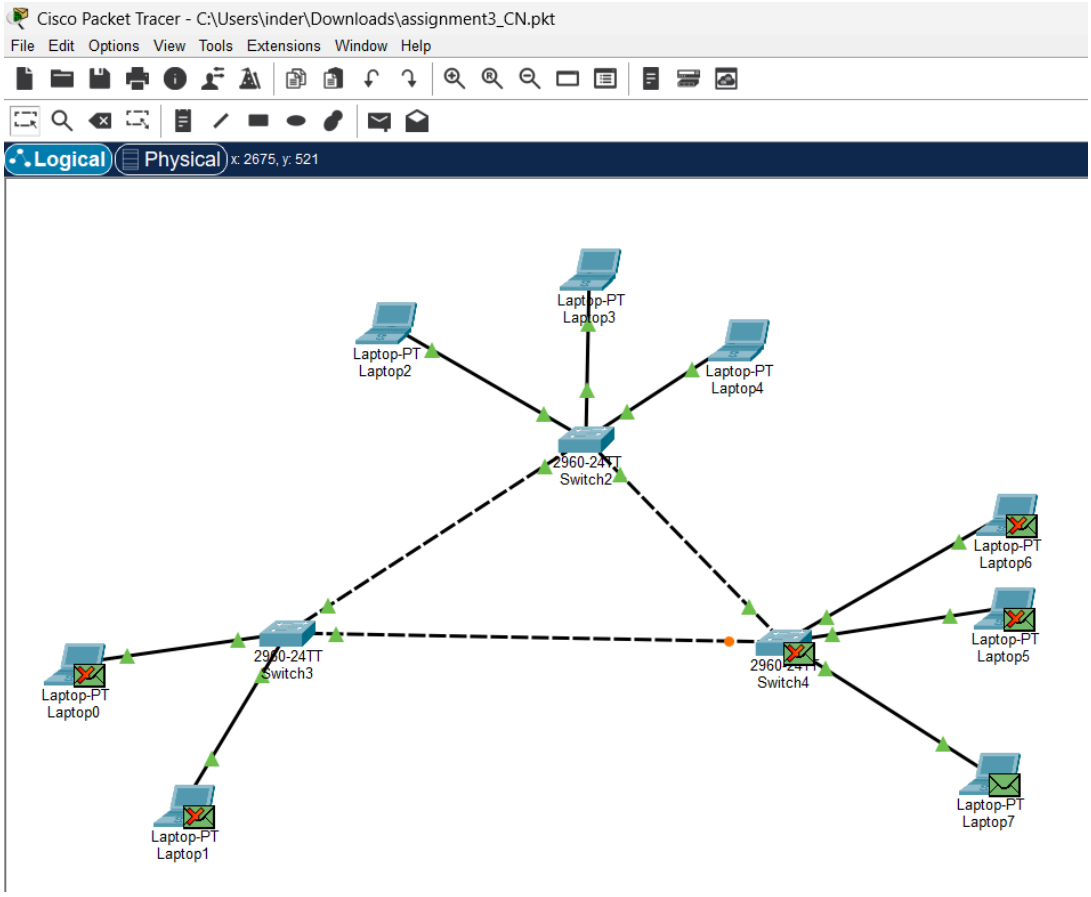
Q6. Create the following Ring Topology.

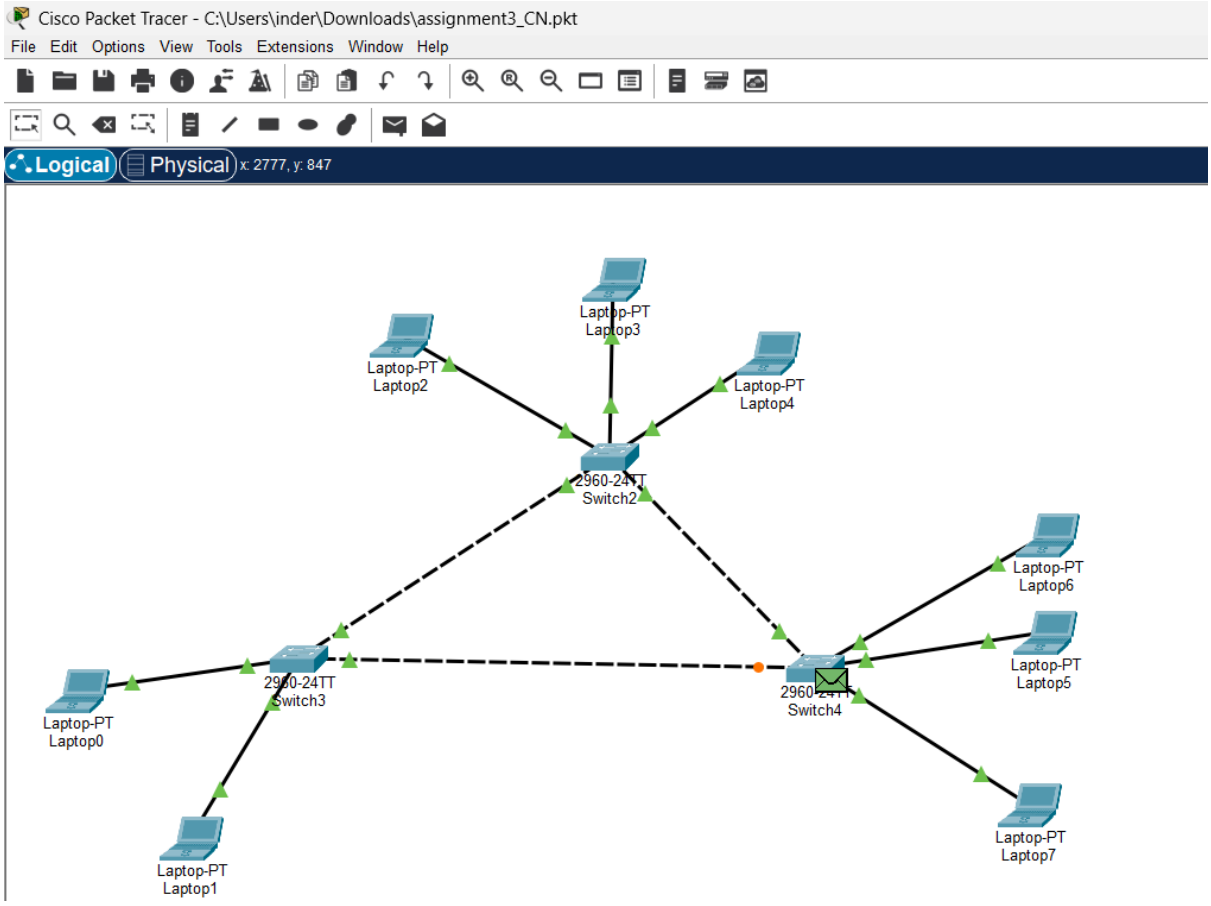


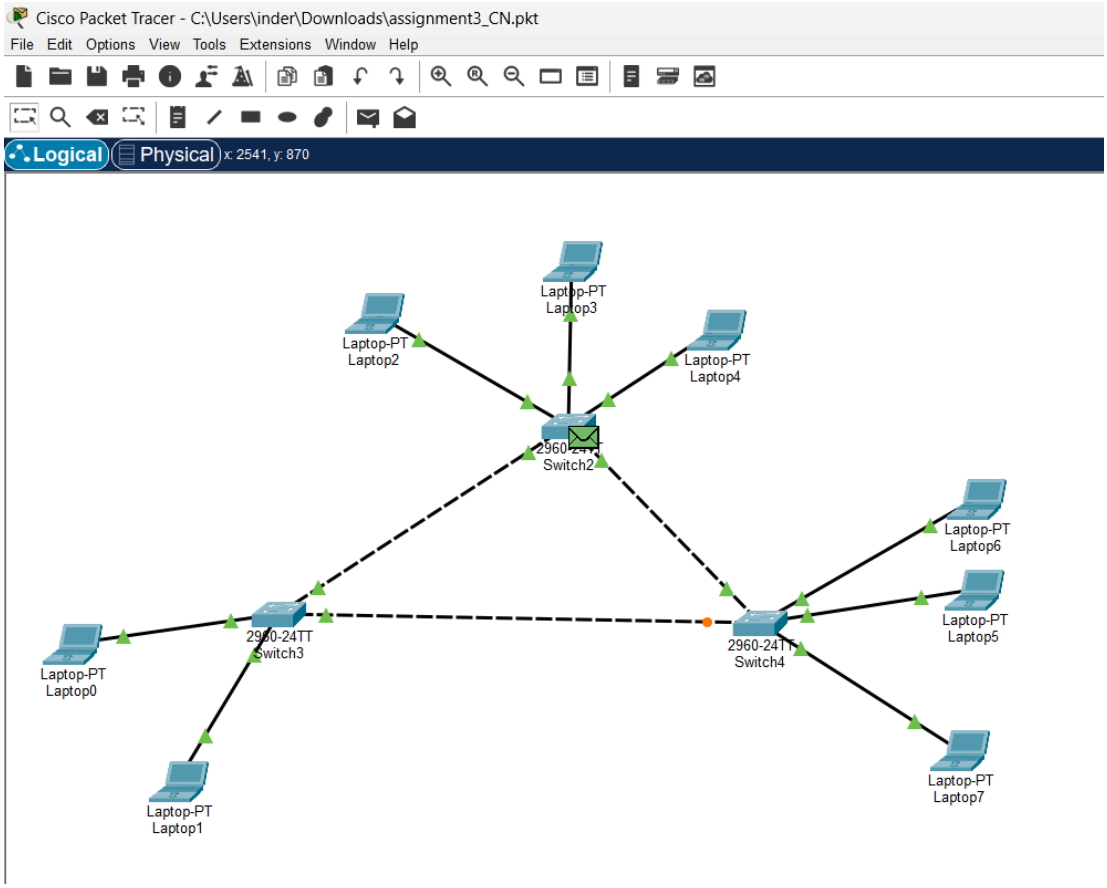
Pinging Laptop7 (192.168.1.36) from Laptop3

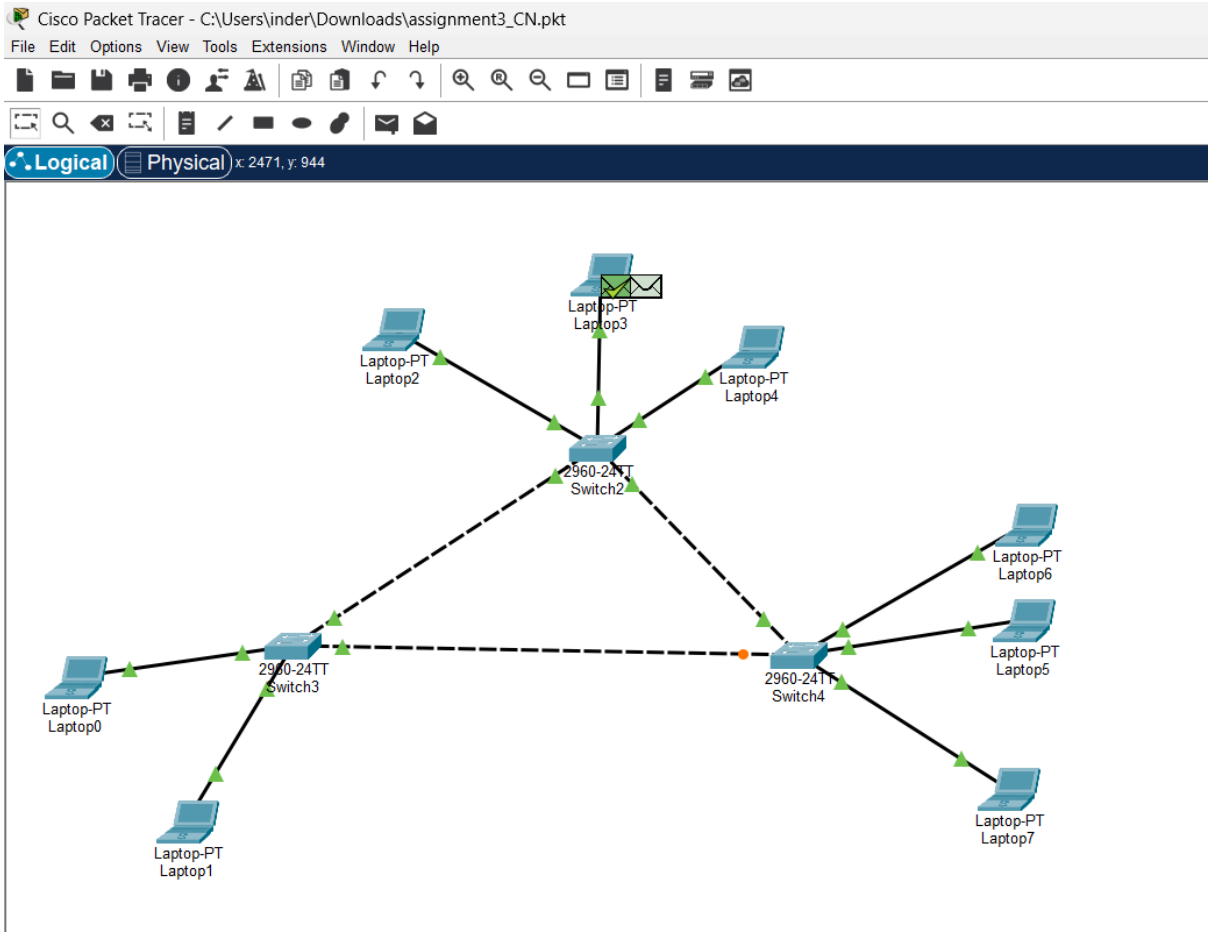

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2727, y: 542
Laptop-PT Laptop3
Laptop-PT Laptop2
Laptop-PT Laptop4
Switch2
Laptop-PT Laptop6
Laptop-PT Laptop5
Switch3
Switch4
Laptop-PT Laptop0
Laptop-PT Laptop7
Laptop-PT Laptop1


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
le Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2475, y: 433
Laptop-PT Laptop3
Laptop-PT Laptop2
Laptop-PT Laptop4
Switch2
Laptop-PT Laptop6
Laptop-PT Laptop5
Switch3
Switch4
Laptop-PT Laptop0
Laptop-PT Laptop7
Laptop-PT Laptop1


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2675, y: 521
Laptop-PT
Laptop3
Laptop-PT
Laptop2
Laptop-PT
Laptop4
2960-24TT
Switch2
Laptop-PT
Laptop6
Laptop-PT
Laptop5
2960-24TT
Switch3
2960-24TT
Switch4
Laptop-PT
Laptop0
Laptop-PT
Laptop7
Laptop-PT
Laptop1


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2777, y: 847
Laptop-PT
Laptop3
Laptop-PT
Laptop2
Laptop-PT
Laptop4
2960-24TT
Switch2
Laptop-PT
Laptop6
Laptop-PT
Laptop5
2960-24TT
Switch3
2960-24TT
Switch4
Laptop-PT
Laptop0
Laptop-PT
Laptop7
Laptop-PT
Laptop1


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2541, y: 870
Laptop-PT Laptop3
Laptop-PT Laptop2
Laptop-PT Laptop4
Switch2
Laptop-PT Laptop6
Laptop-PT Laptop5
2960-24TT Switch3
2960-24TT Switch4
Laptop-PT Laptop0
Laptop-PT Laptop7
Laptop-PT Laptop1


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 2471, y: 944
Laptop-PT Laptop3
Laptop-PT Laptop2
Laptop-PT Laptop4
Switch2
Laptop-PT Laptop6
Laptop-PT Laptop5
2960-24TT Switch3
2960-24TT Switch4
Laptop-PT Laptop0
Laptop-PT Laptop7
Laptop-PT Laptop1

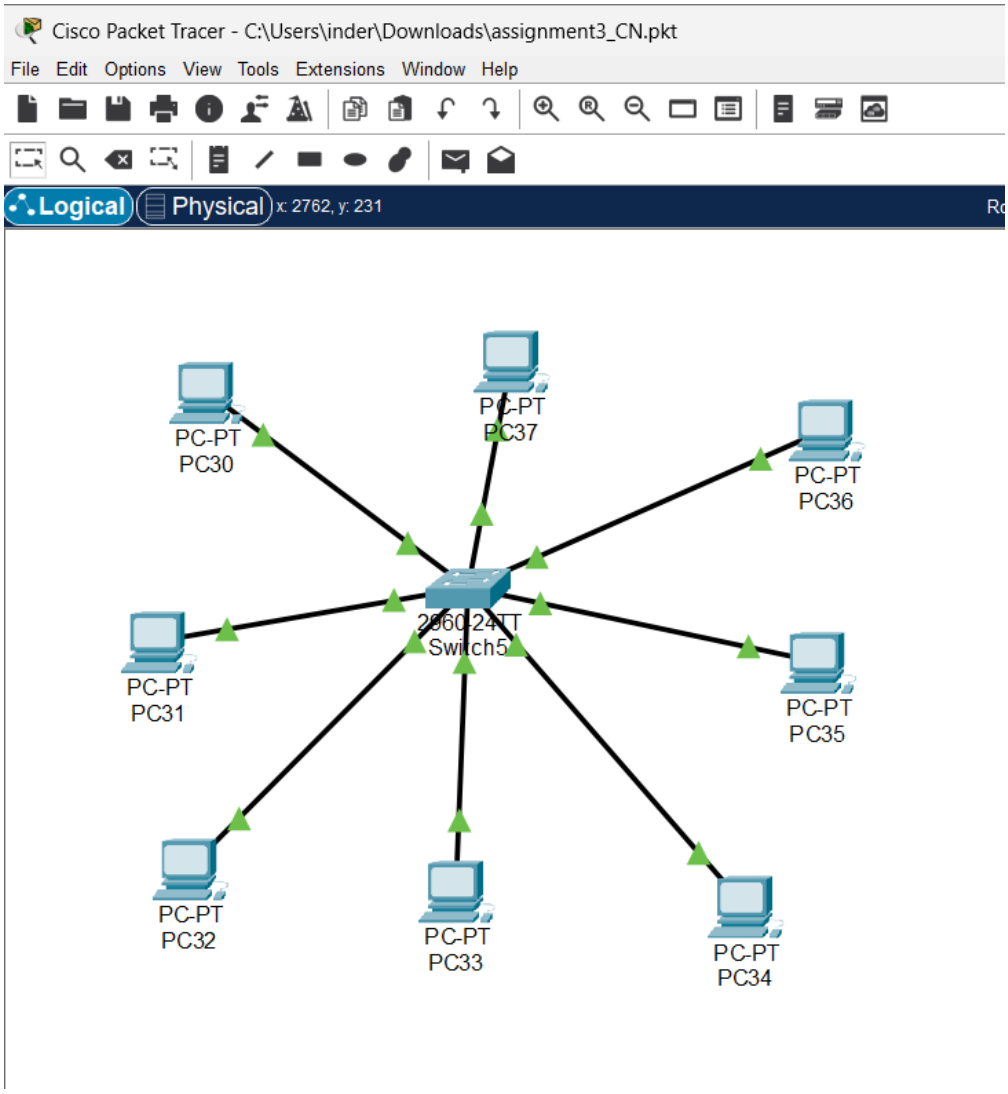
Laptop3

Physical Config Desktop Programming Attributes

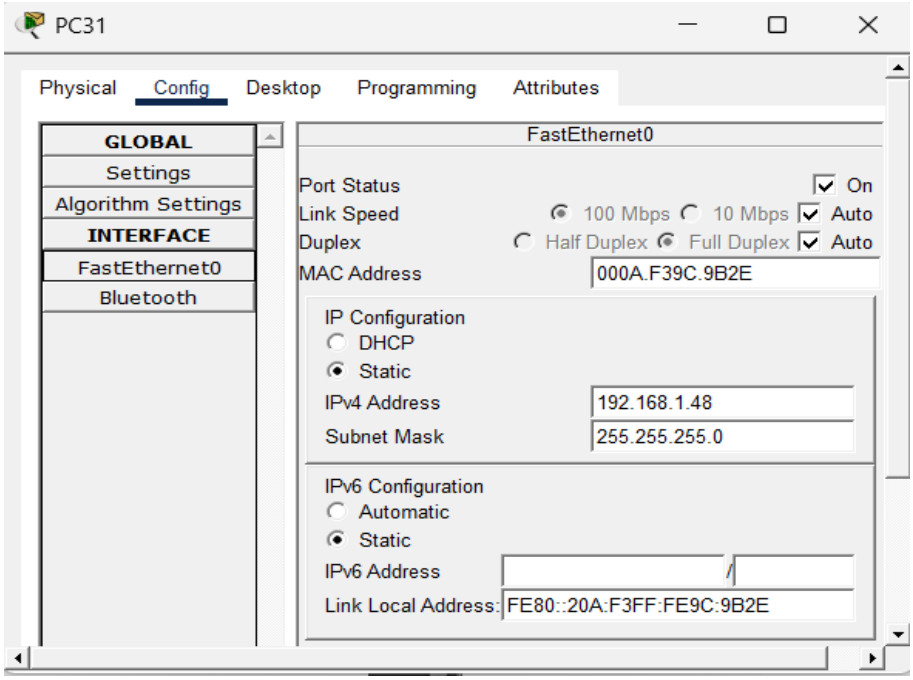
Command Prompt

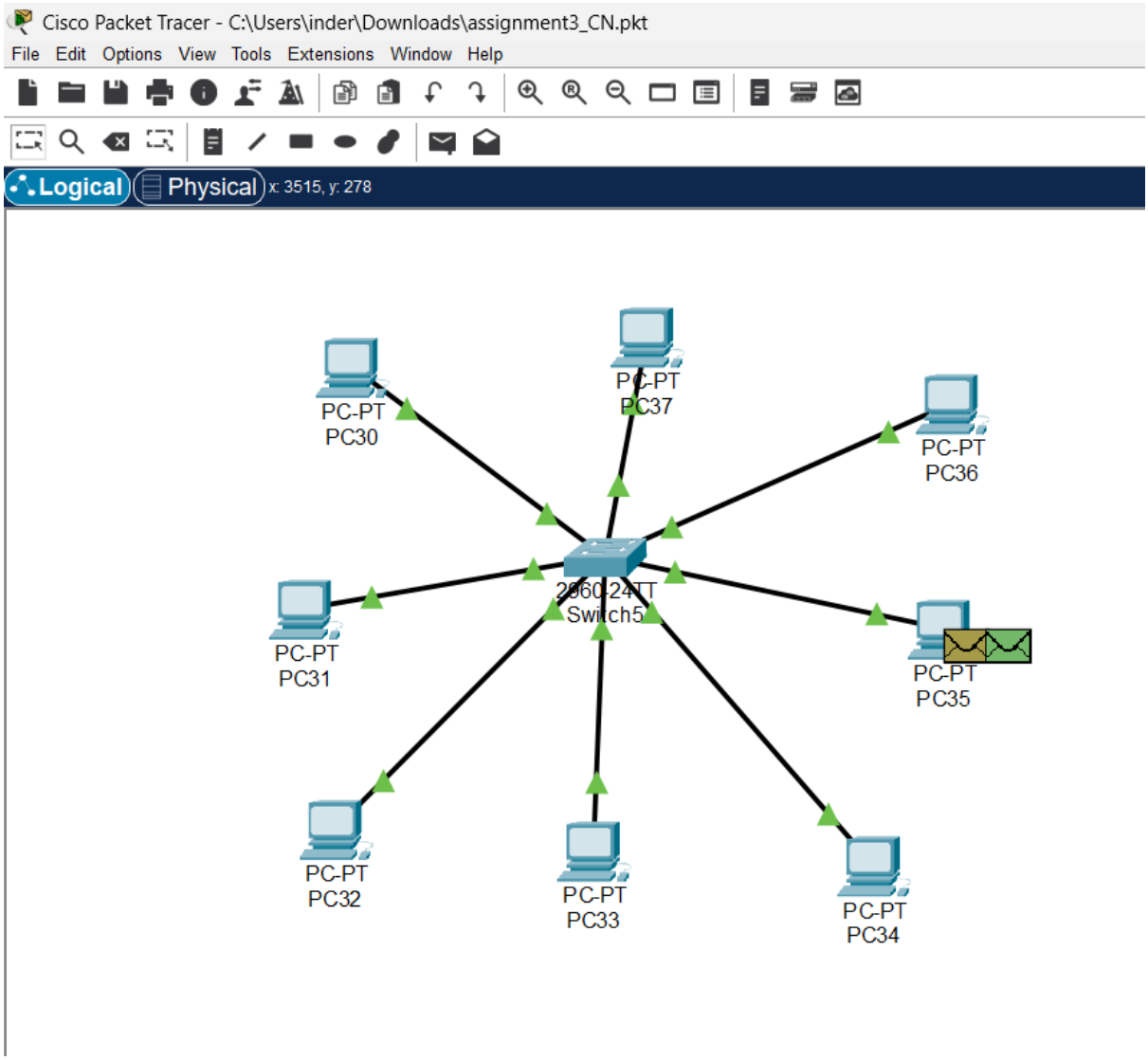
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.36

Pinging 192.168.1.36 with 32 bytes of data:

Reply from 192.168.1.36: bytes=32 time=12ms TTL=128
Reply from 192.168.1.36: bytes=32 time=26ms TTL=128
Reply from 192.168.1.36: bytes=32 time<1ms TTL=128
Reply from 192.168.1.36: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 26ms, Average = 9ms

C:\>
```

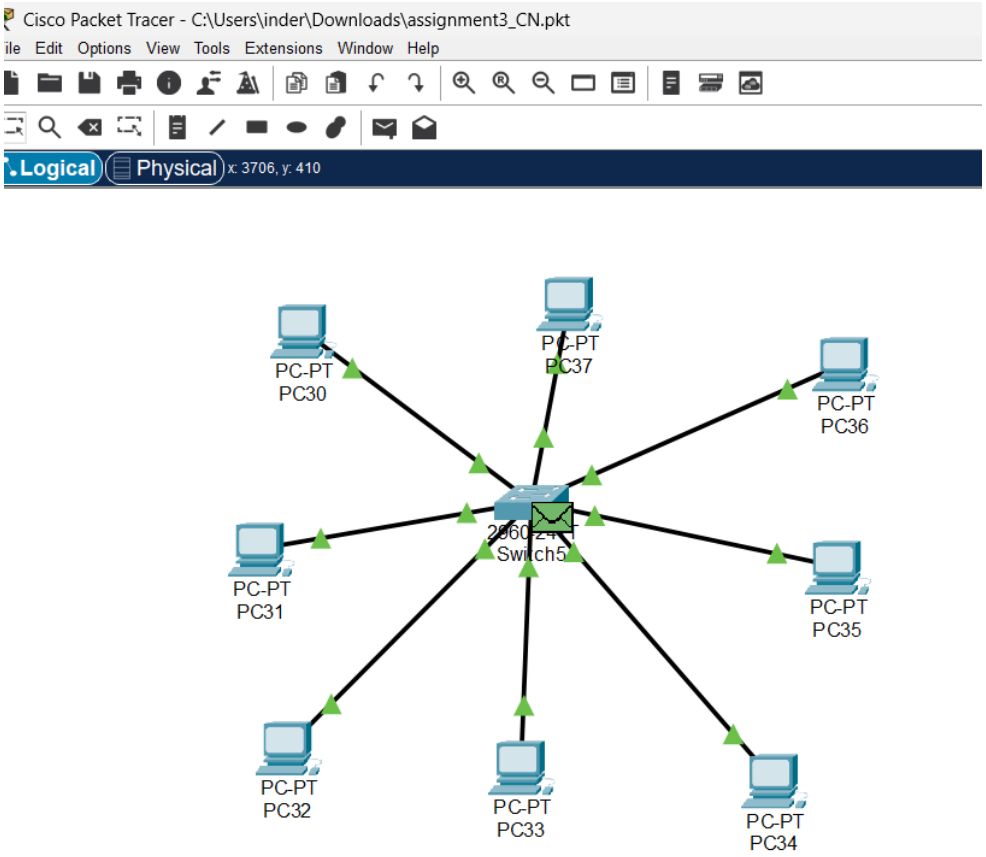
Q7. Create the following Ring Topology.

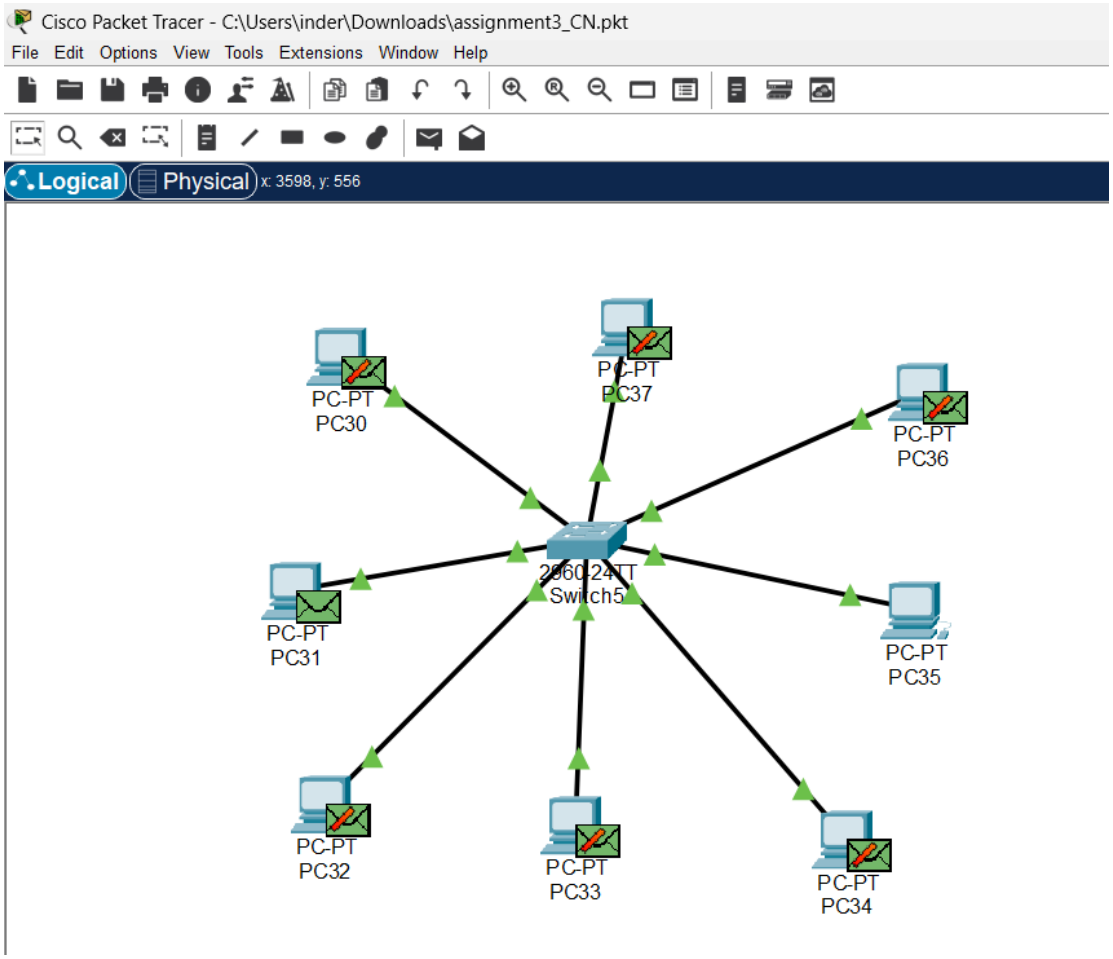


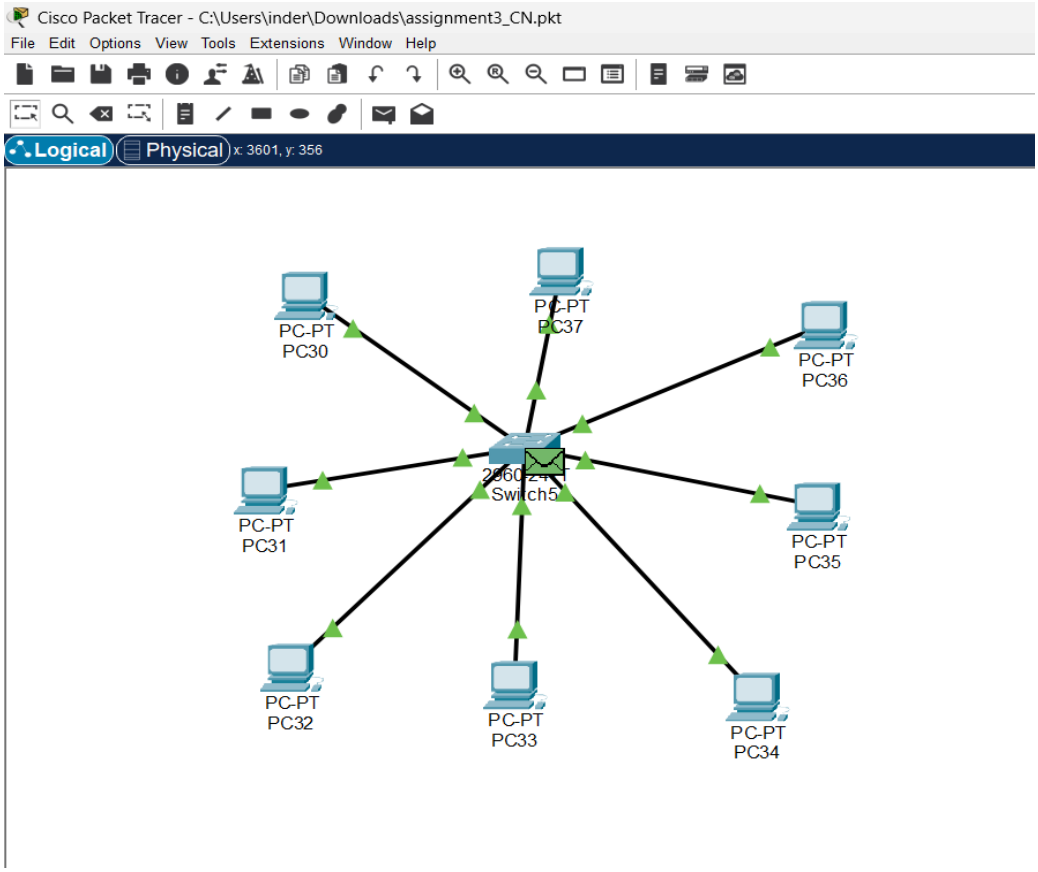
Pinging PC31(192.168.1.48) from PC35

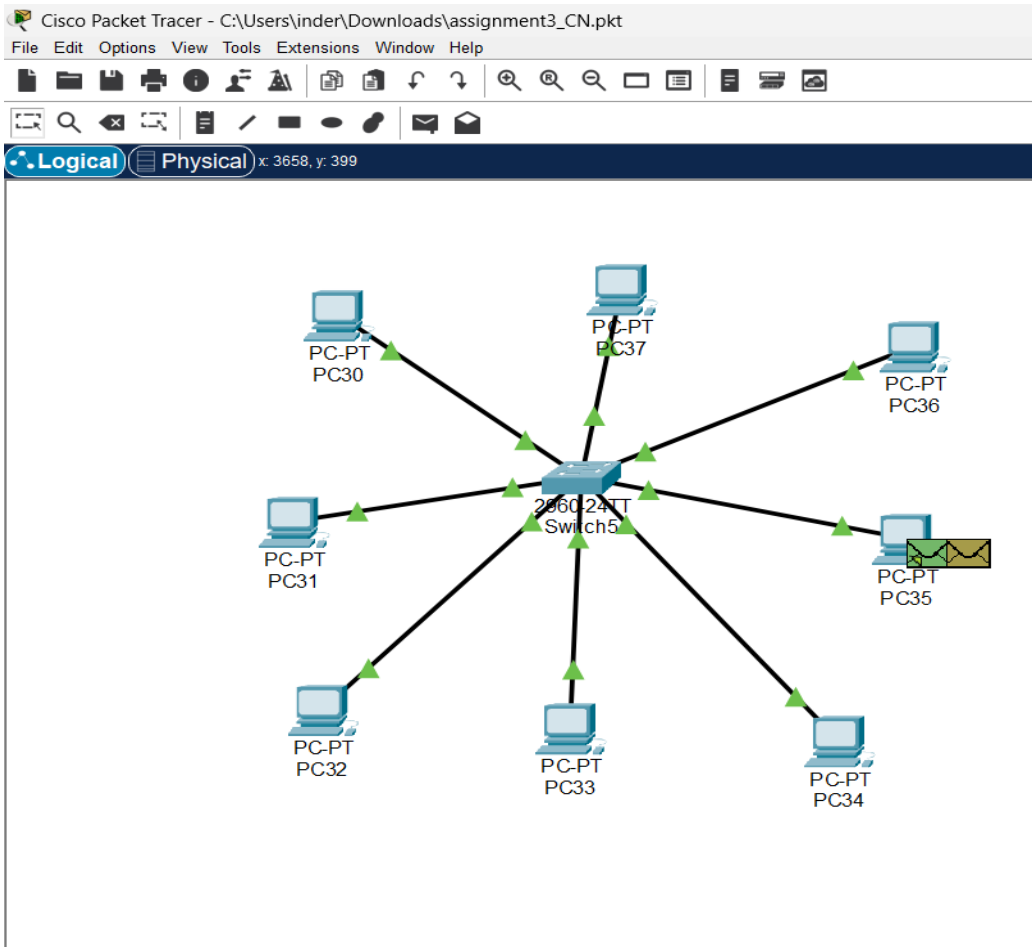

PC31
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000A.F39C.9B2E
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.48
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::20A:F3FF:FE9C:9B2E

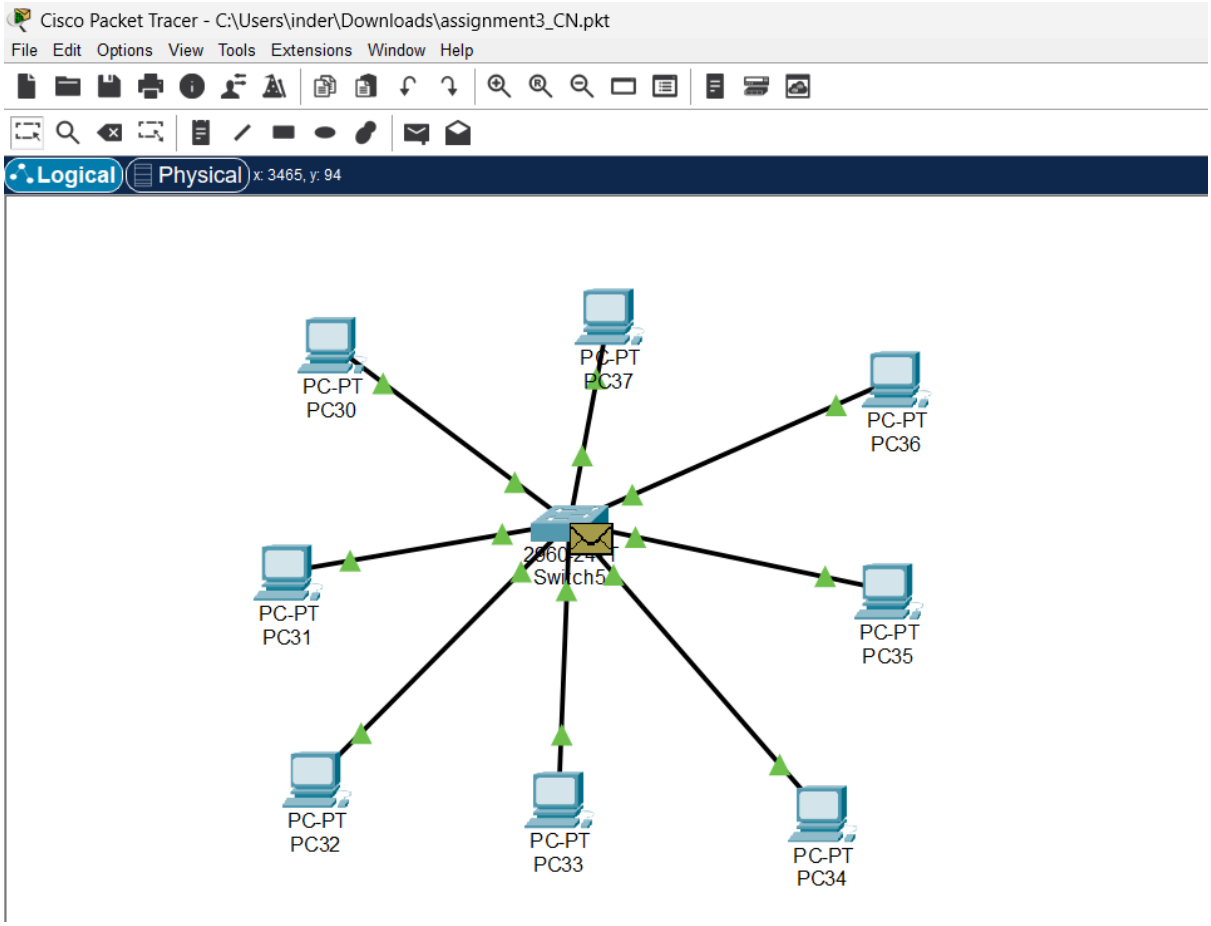

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3515, y: 278
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
2960-24TT
Switch5
PC-PT
PC31
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

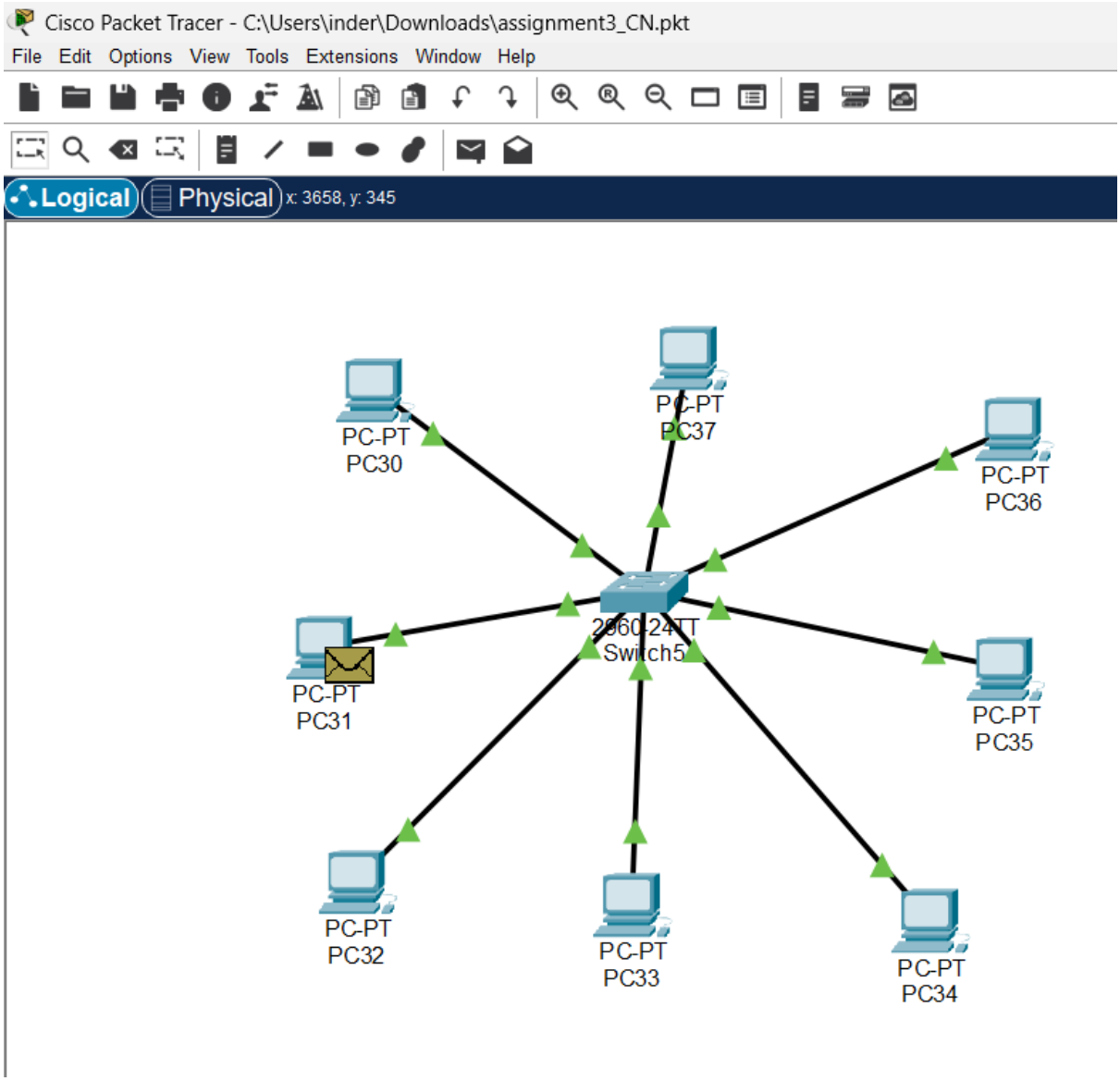

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
ile Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3706, y: 410
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
PC-PT
PC31
Switch5
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

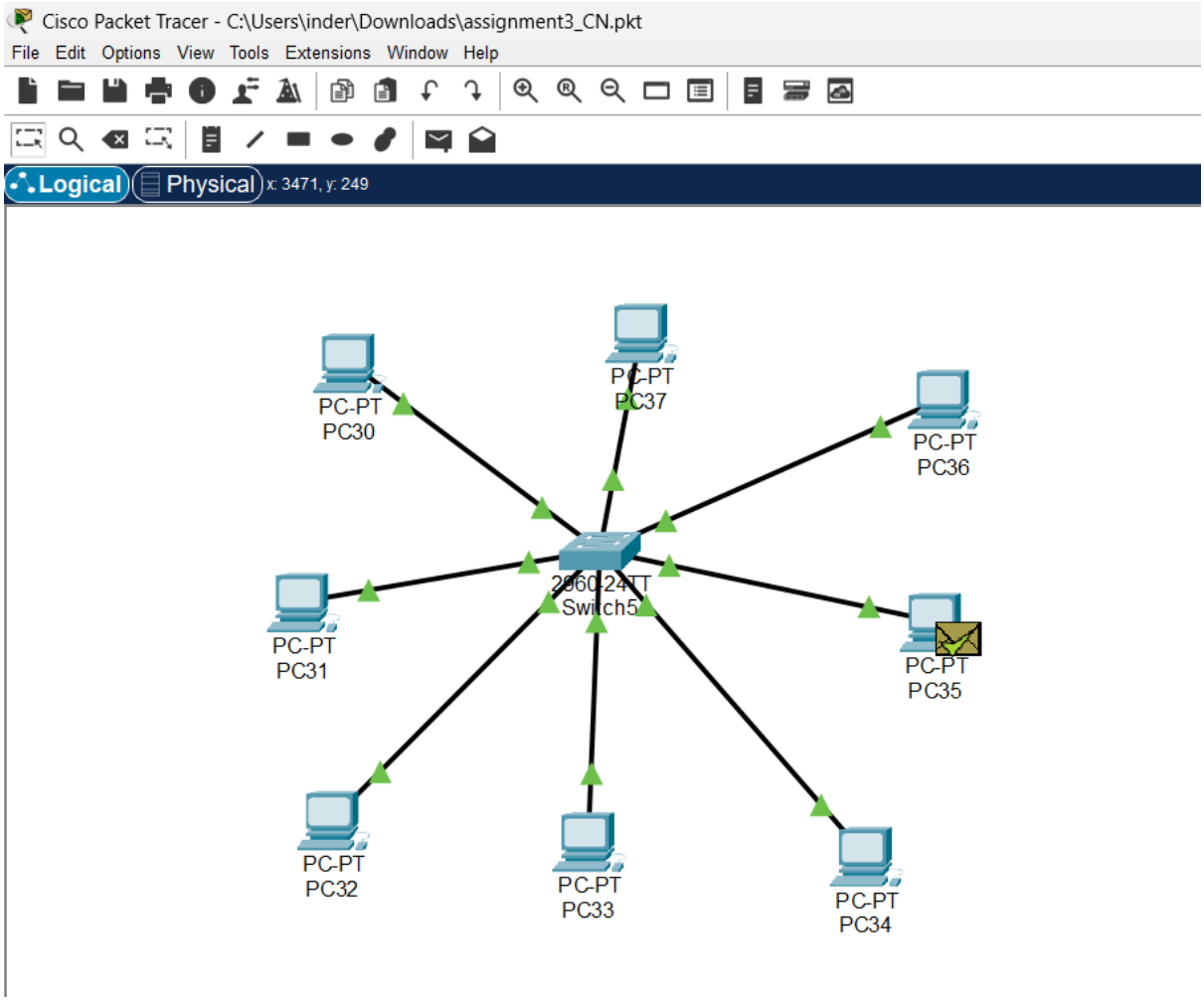

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3598, y: 556
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
PC-PT
PC31
Switch5
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

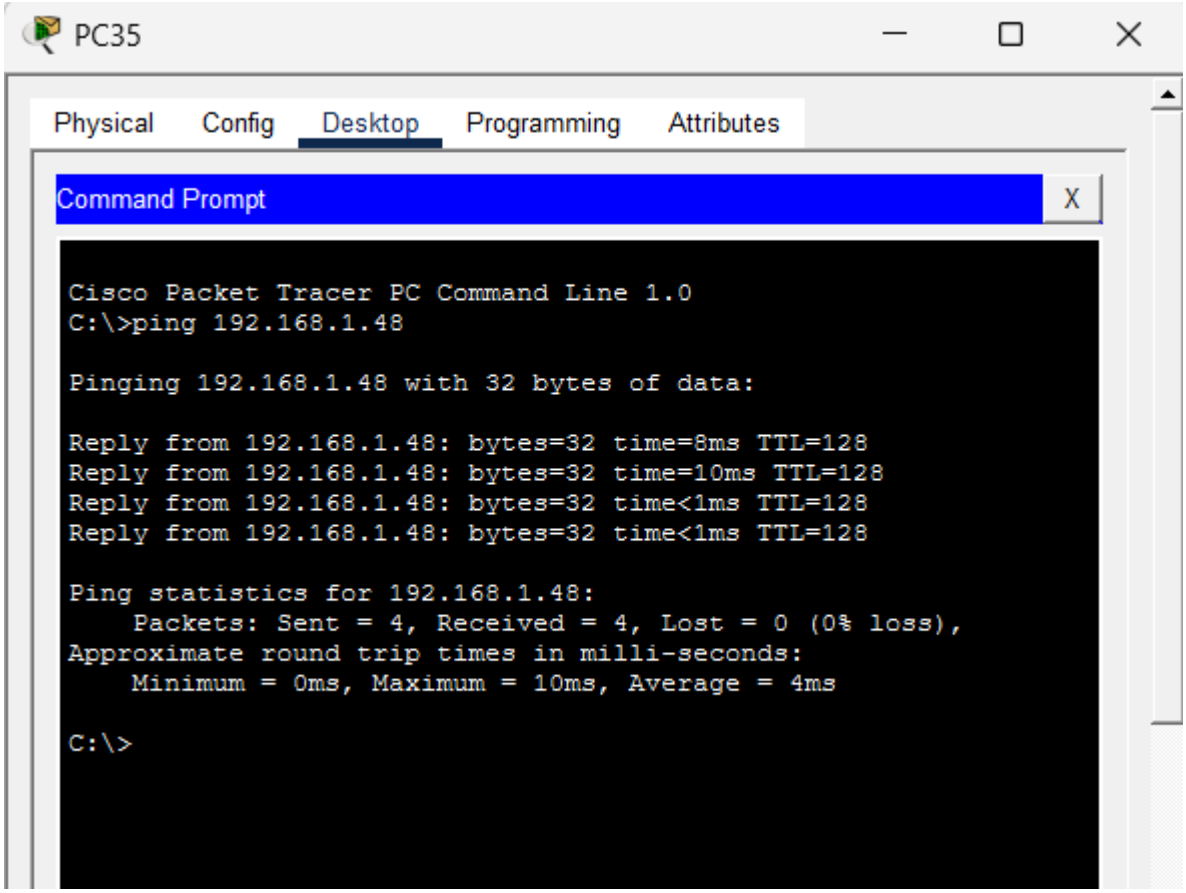

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3601, y: 356
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
PC-PT
PC31
Switch5
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3658, y: 399
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
PC-PT
PC31
Switch5
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

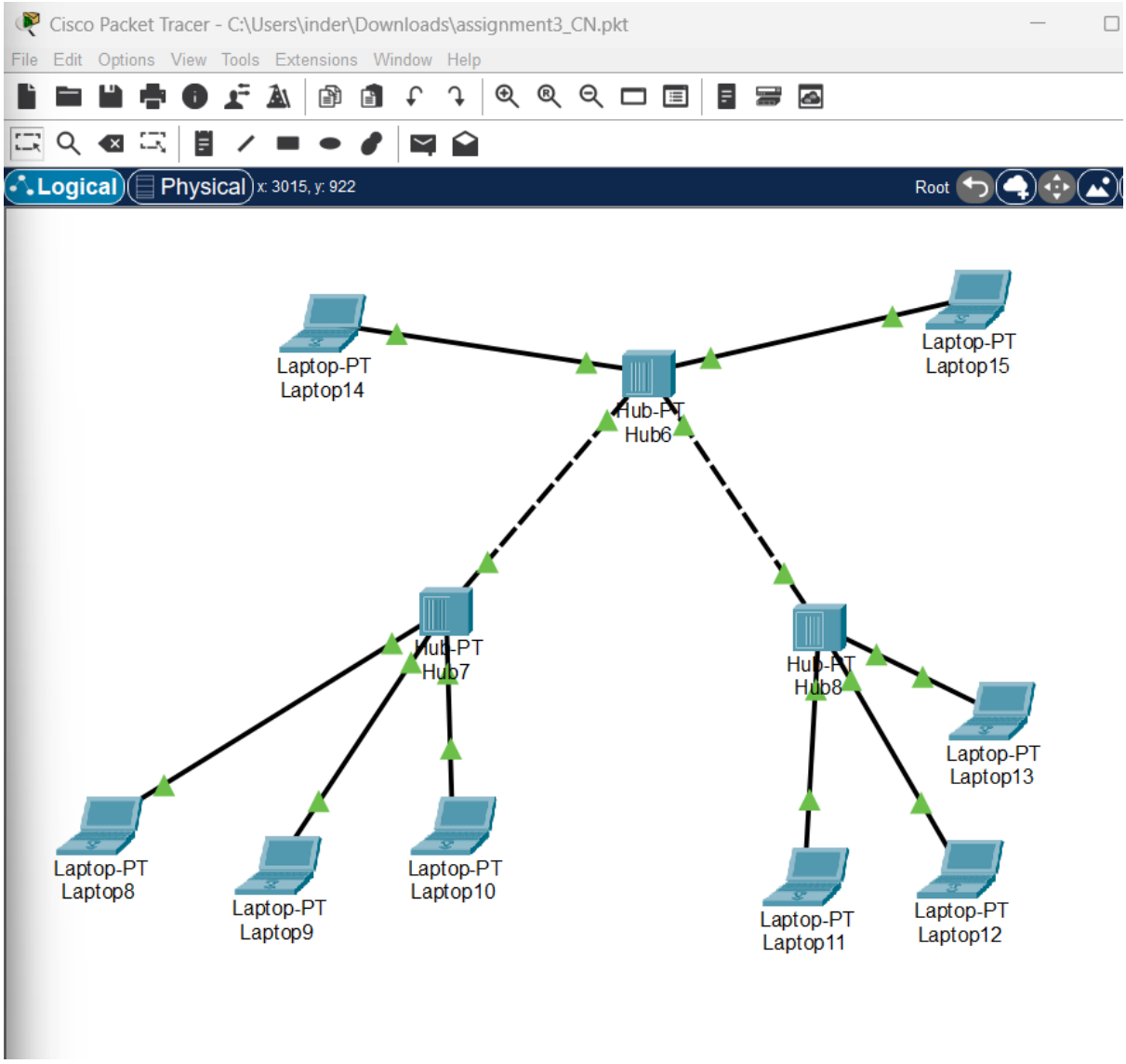

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3465, y: 94
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
PC-PT
PC31
Switch5
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

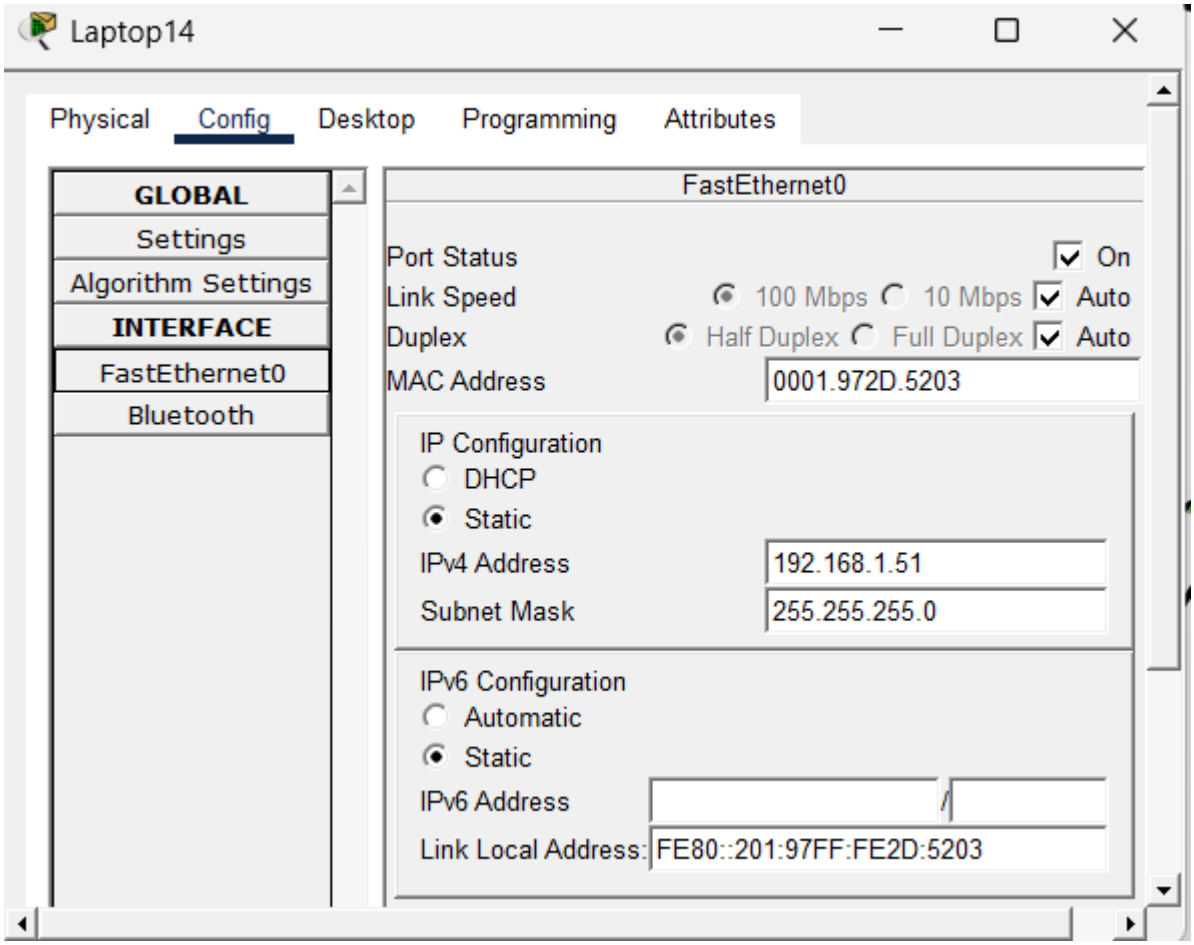

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3658, y: 345
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
2960-24TT
Switch5
PC-PT
PC31
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34

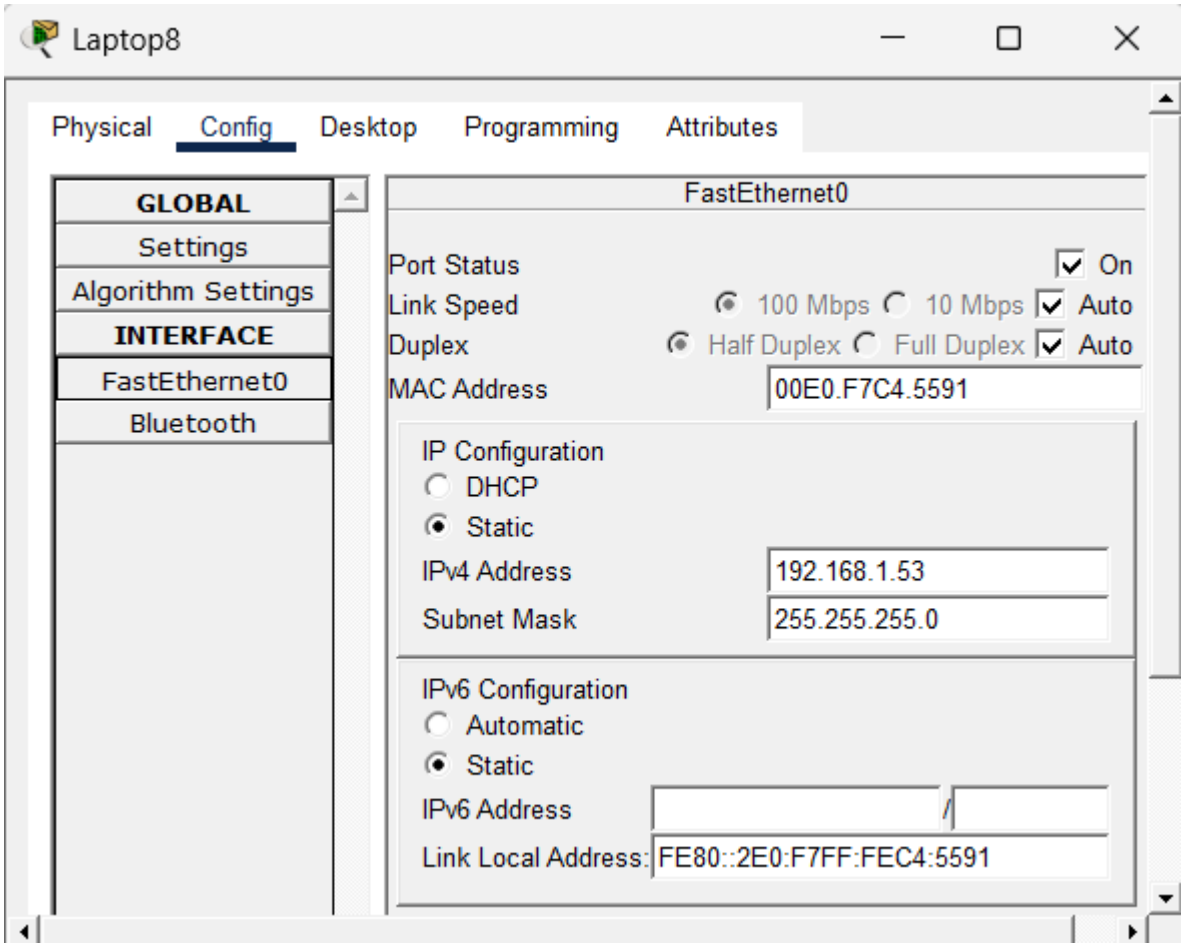

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3471, y: 249
PC-PT
PC30
PC-PT
PC37
PC-PT
PC36
2960-24TT
Switch5
PC-PT
PC31
PC-PT
PC35
PC-PT
PC32
PC-PT
PC33
PC-PT
PC34



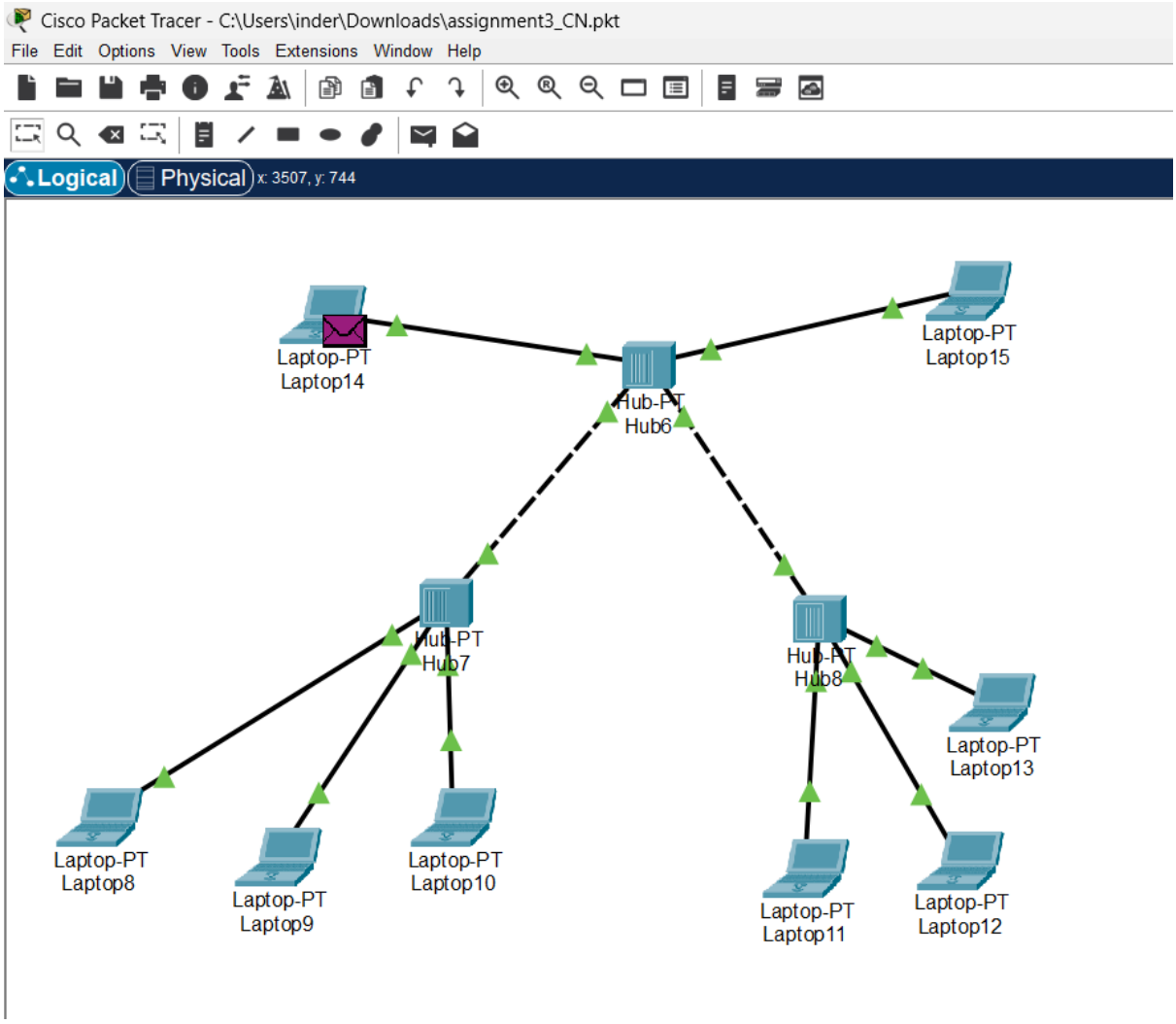
Q8. Create the following Tree Topology.

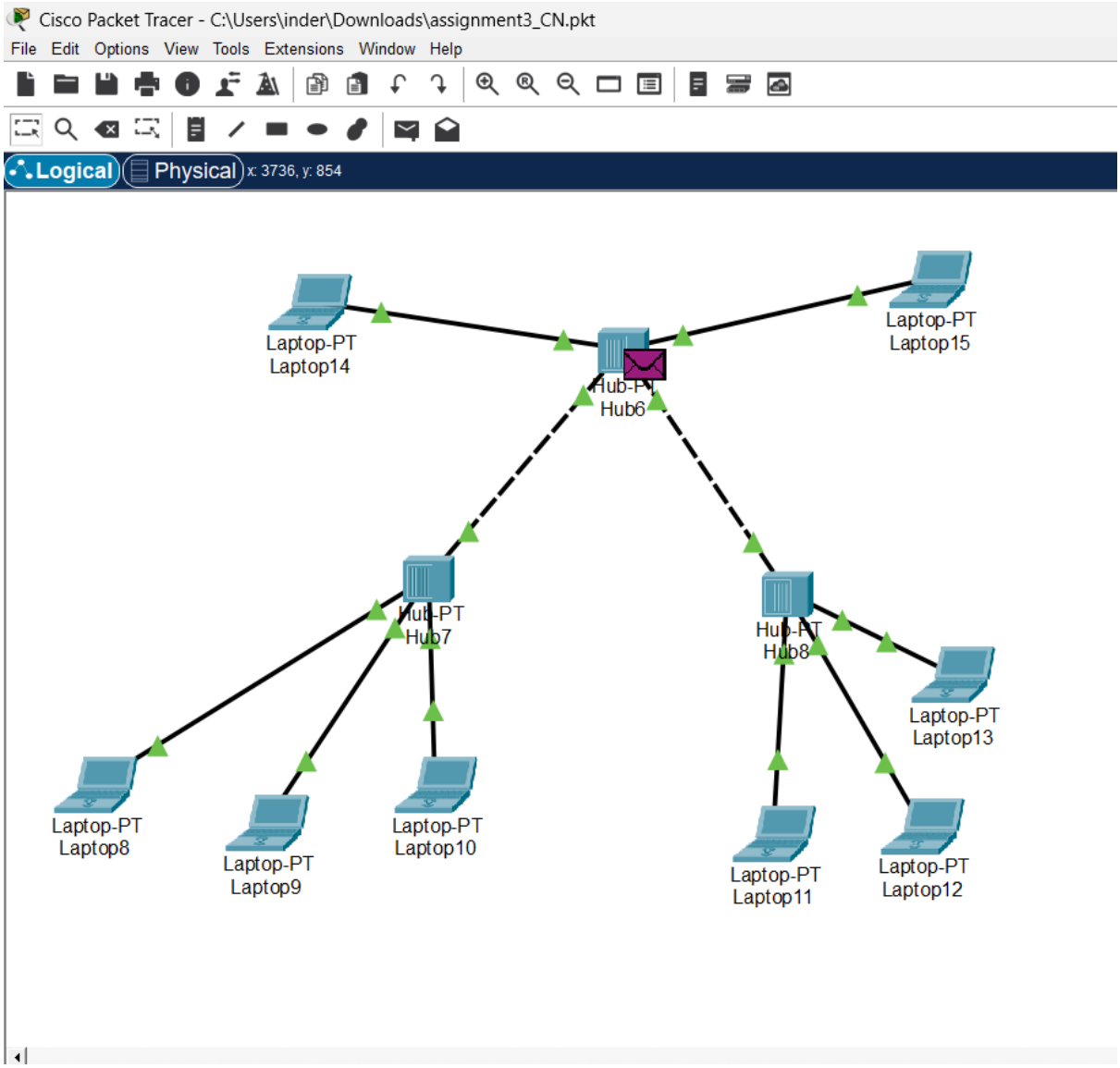

Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3015, y: 922
Root
Laptop-PT
Laptop14
Laptop-PT
Laptop15
Hub-PT
Hub6
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12

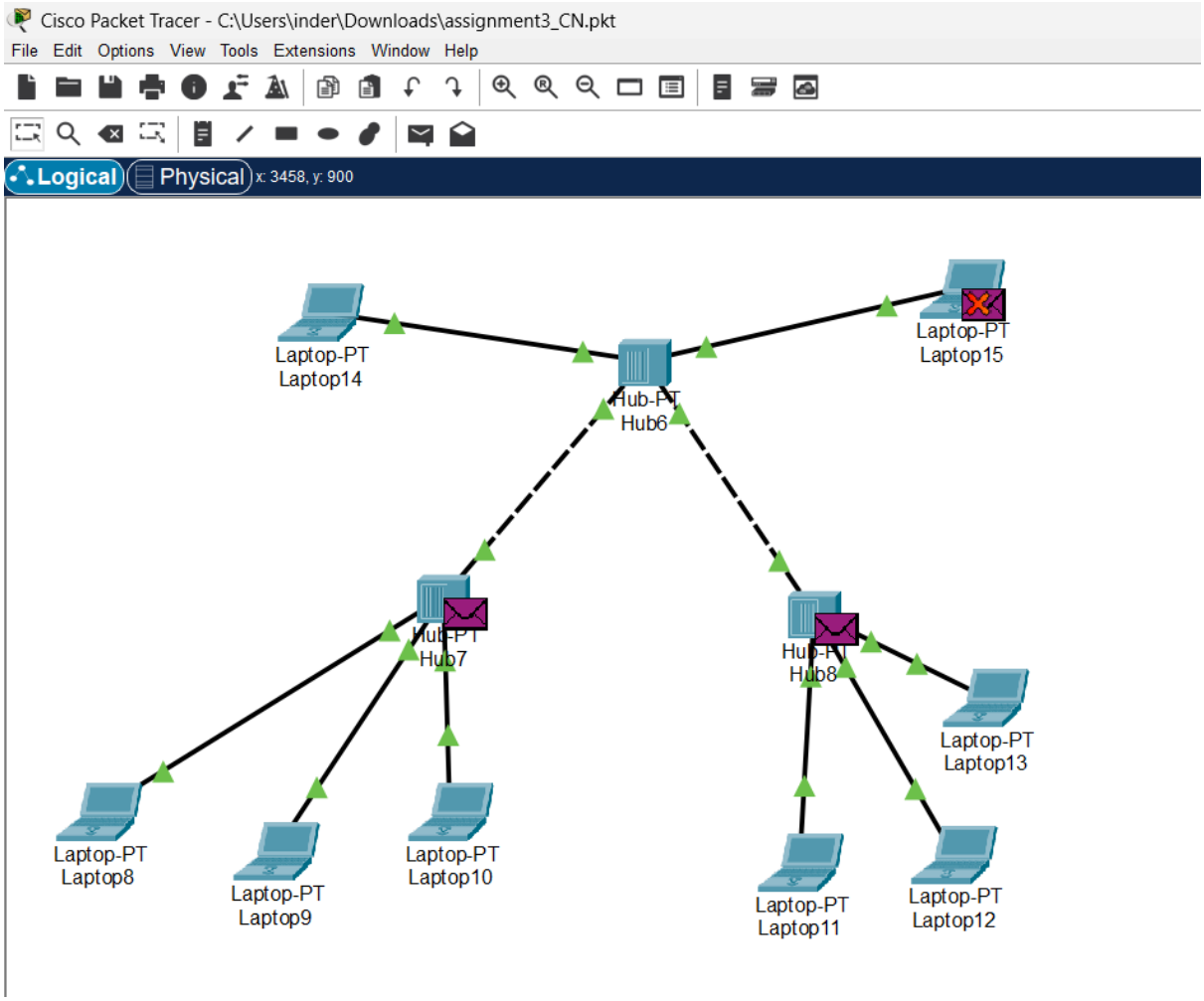

Laptop14
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.972D.5203
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.51
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::201:97FF:FE2D:5203

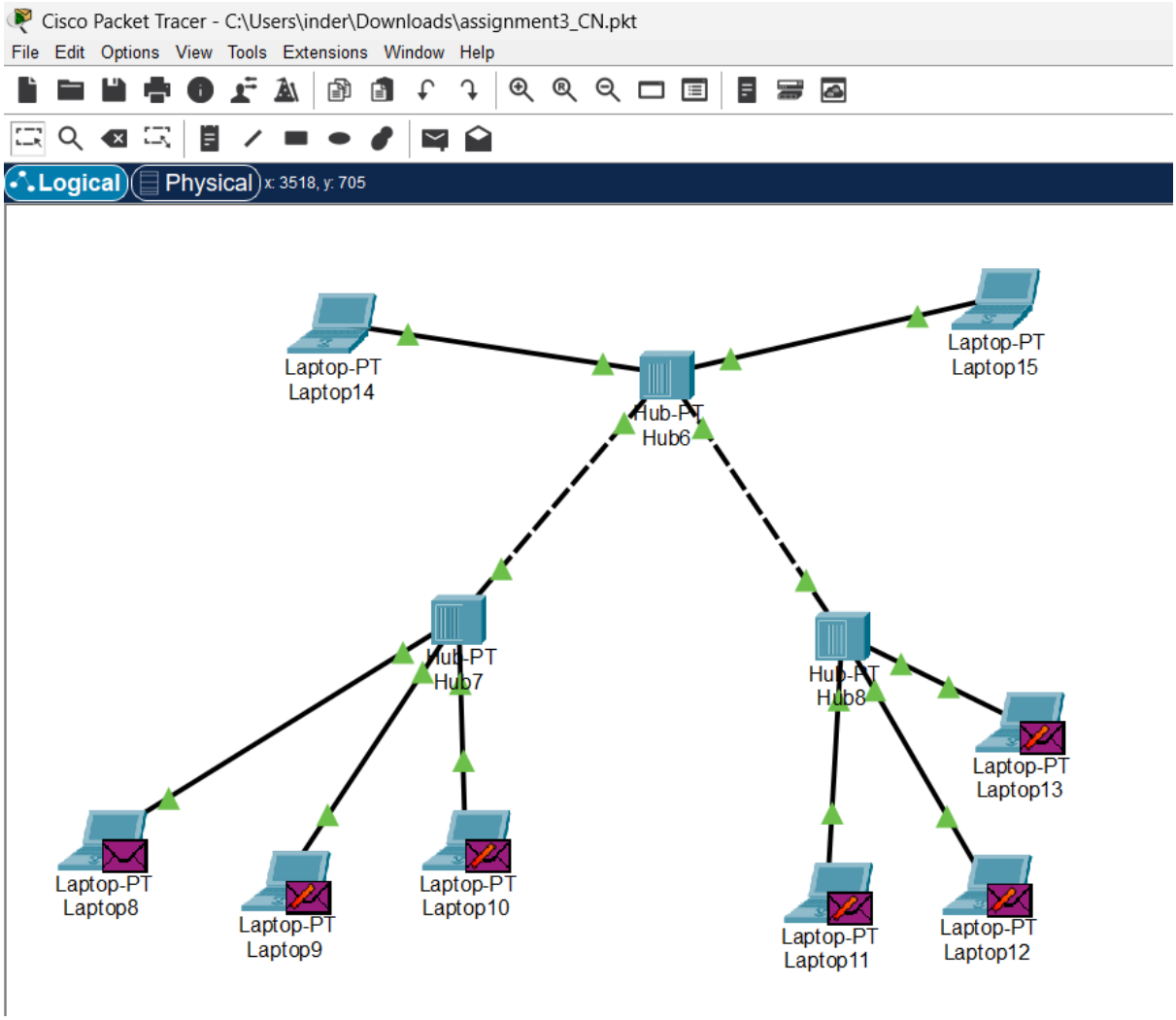


Pinging Laptop8 (192.168.1.53) from Laptop14 (192.168.1.51)


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3507, y: 744
Laptop-PT
Laptop14
Hub-PT
Hub6
Laptop-PT
Laptop15
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3736, y: 854
Laptop-PT
Laptop14
Laptop-PT
Laptop15
Hub-PT
Hub6
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3458, y: 900
Laptop-PT
Laptop14
Laptop-PT
Laptop15
Hub-PT
Hub6
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3518, y: 705
Laptop-PT
Laptop14
Laptop-PT
Laptop15
Hub-PT
Hub6
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
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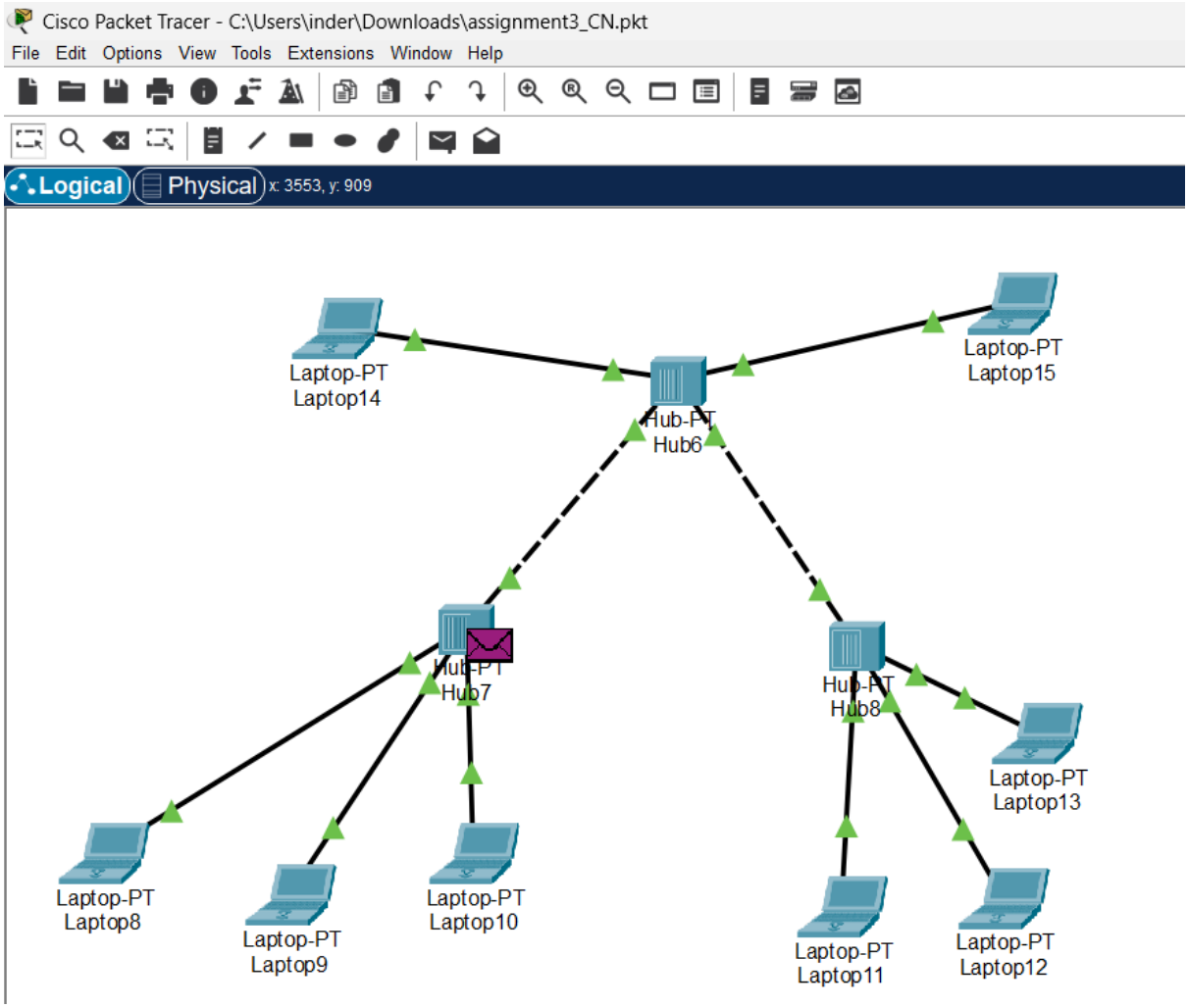
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3553, y: 909
Laptop-PT
Laptop14
Hub-PT
Hub6
Laptop-PT
Laptop15
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
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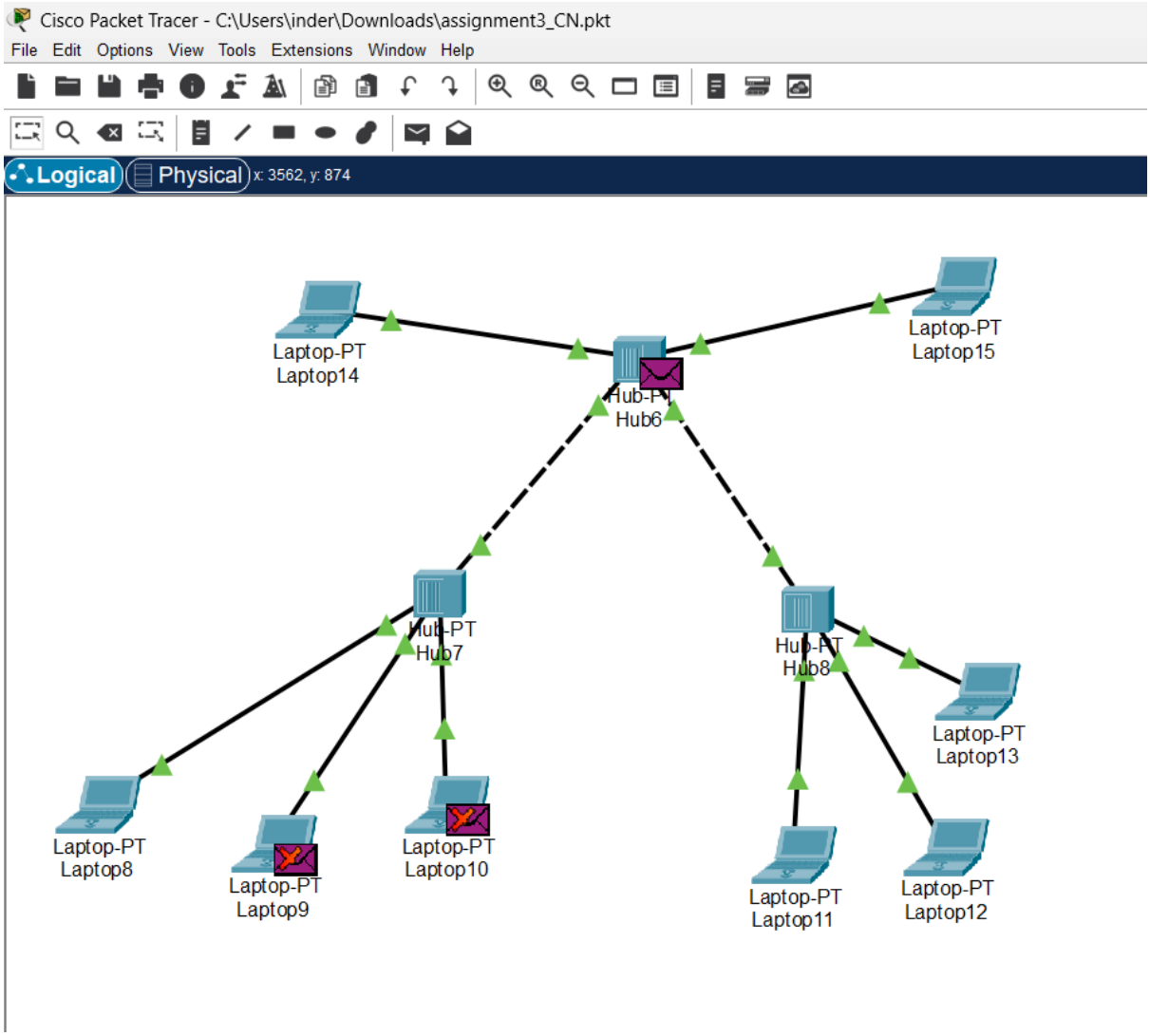
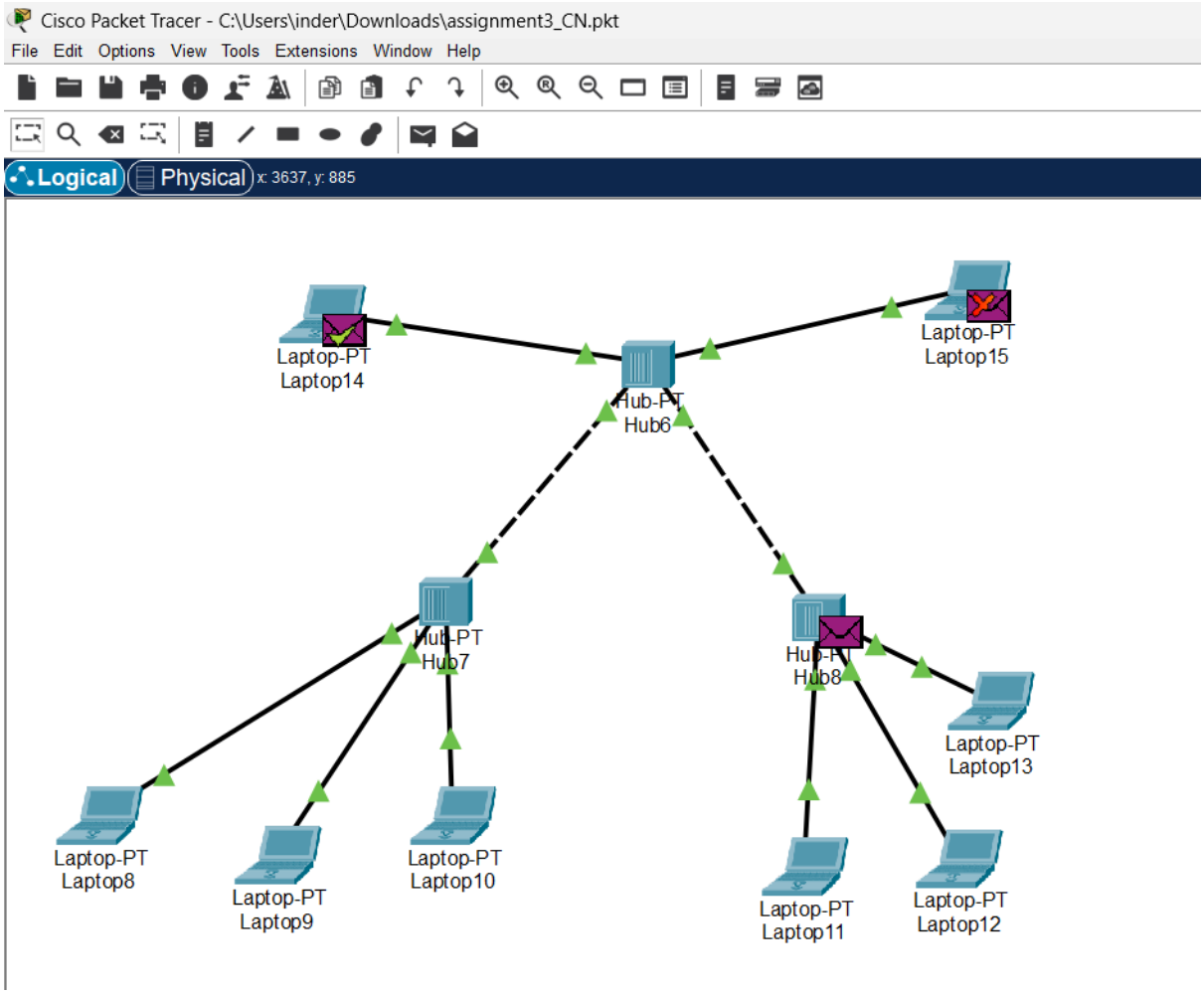
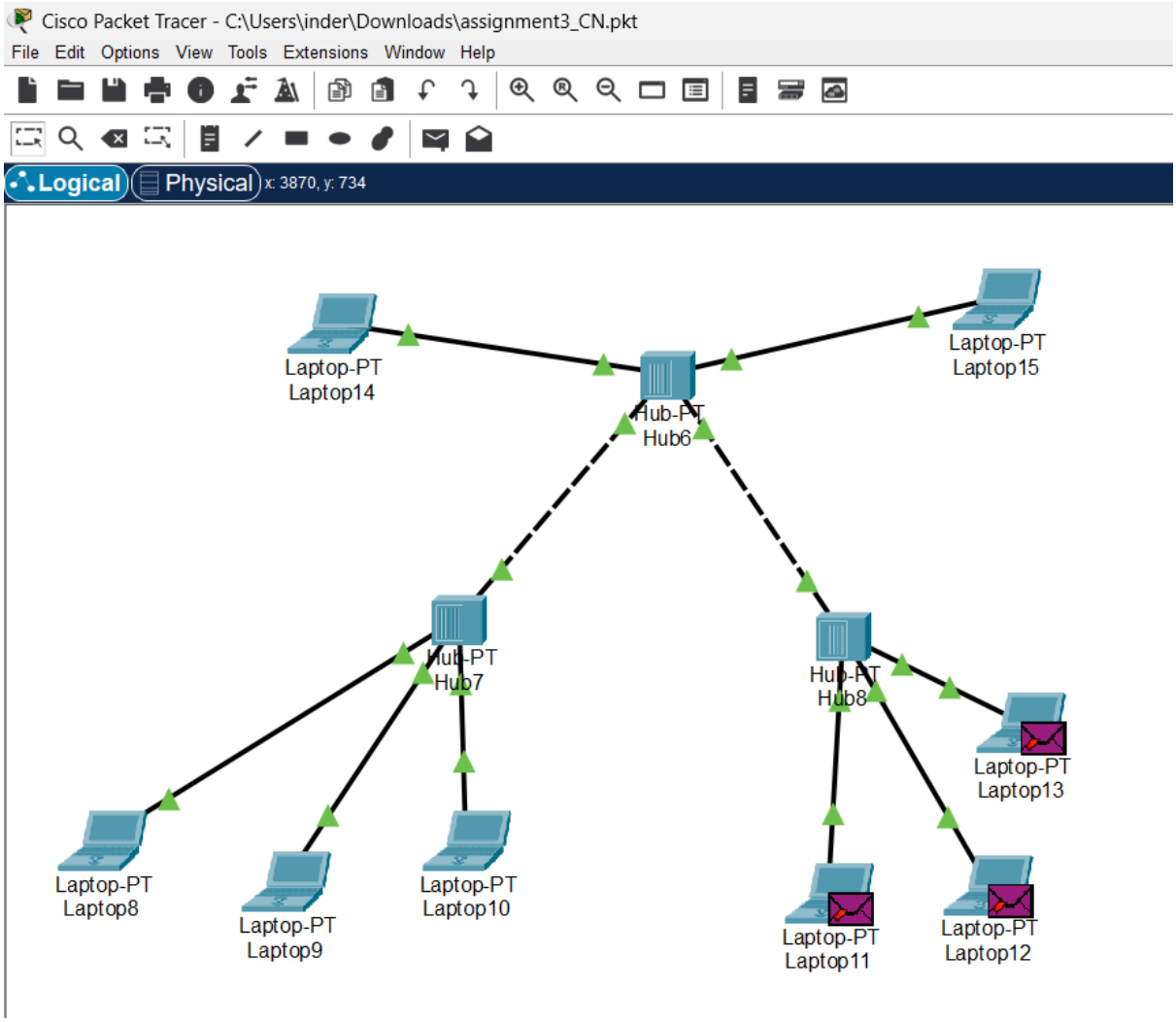
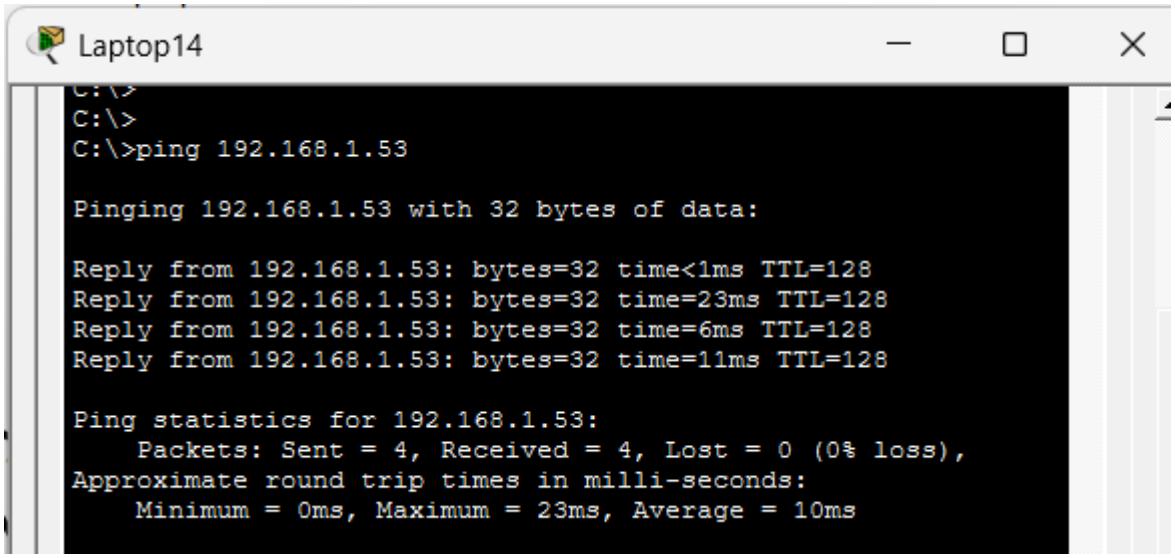
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3562, y: 874
Laptop-PT
Laptop14
Laptop-PT
Laptop15
Hub-PT
Hub6
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3637, y: 885
Laptop-PT
Laptop14
Hub-PT
Hub6
Laptop-PT
Laptop15
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Cisco Packet Tracer - C:\Users\inder\Downloads\assignment3_CN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 3870, y: 734
Laptop-PT
Laptop14
Hub-PT
Hub6
Laptop-PT
Laptop15
Hub-PT
Hub7
Hub-PT
Hub8
Laptop-PT
Laptop13
Laptop-PT
Laptop8
Laptop-PT
Laptop9
Laptop-PT
Laptop10
Laptop-PT
Laptop11
Laptop-PT
Laptop12


Laptop14
C:\>
C:\>
C:\>ping 192.168.1.53
Pinging 192.168.1.53 with 32 bytes of data:
Reply from 192.168.1.53: bytes=32 time<1ms TTL=128
Reply from 192.168.1.53: bytes=32 time=23ms TTL=128
Reply from 192.168.1.53: bytes=32 time=6ms TTL=128
Reply from 192.168.1.53: bytes=32 time=11ms TTL=128
Ping statistics for 192.168.1.53:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 23ms, Average = 10ms